
‘Omics powered by Nanopore-based sensing

Dr Spike Willcocks

Chief Strategy Officer & Co-Founder

Oxford Nanopore Technologies (Est. 9th March 2005)

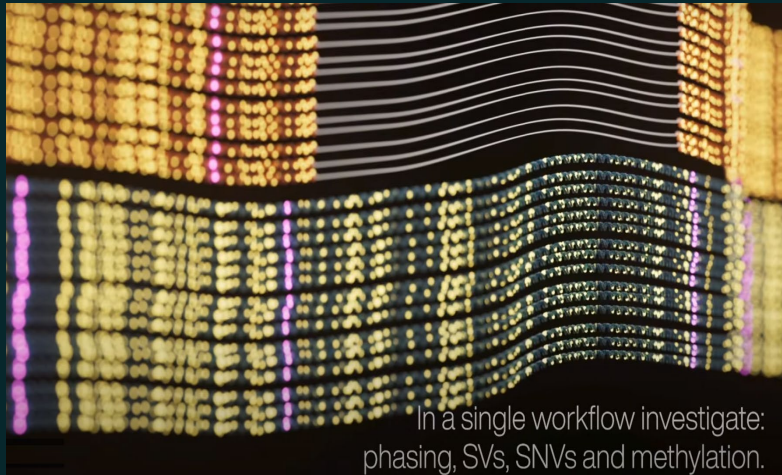


Our nanopore-based sensing platform

“Anything, by anyone, anywhere”



Richer insights



In a single workflow investigate:
phasing, SVs, SNVs and methylation.



Faster time to result



Complement rapid sample prep with
data streaming & no need for batching



Accessible & affordable

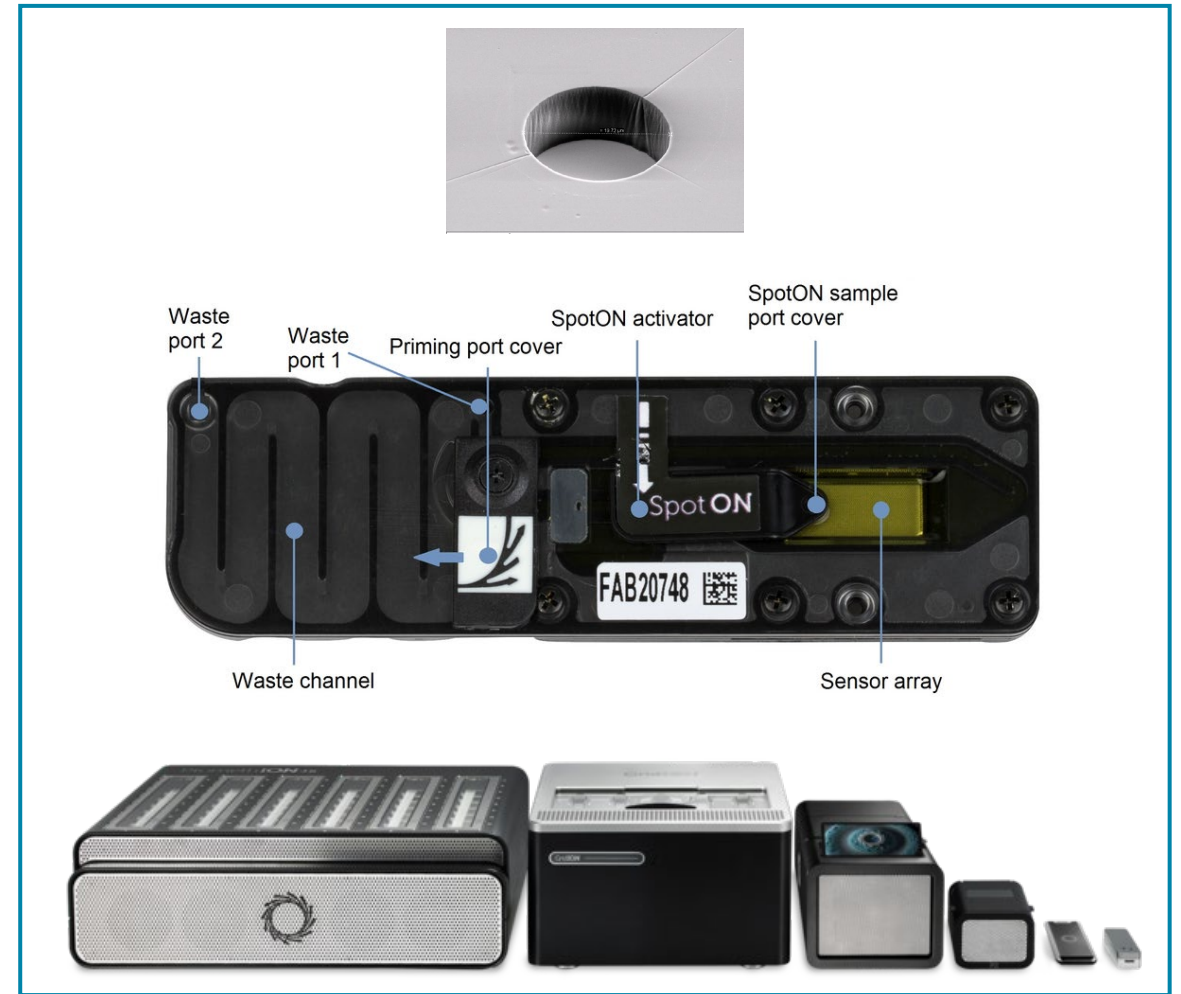
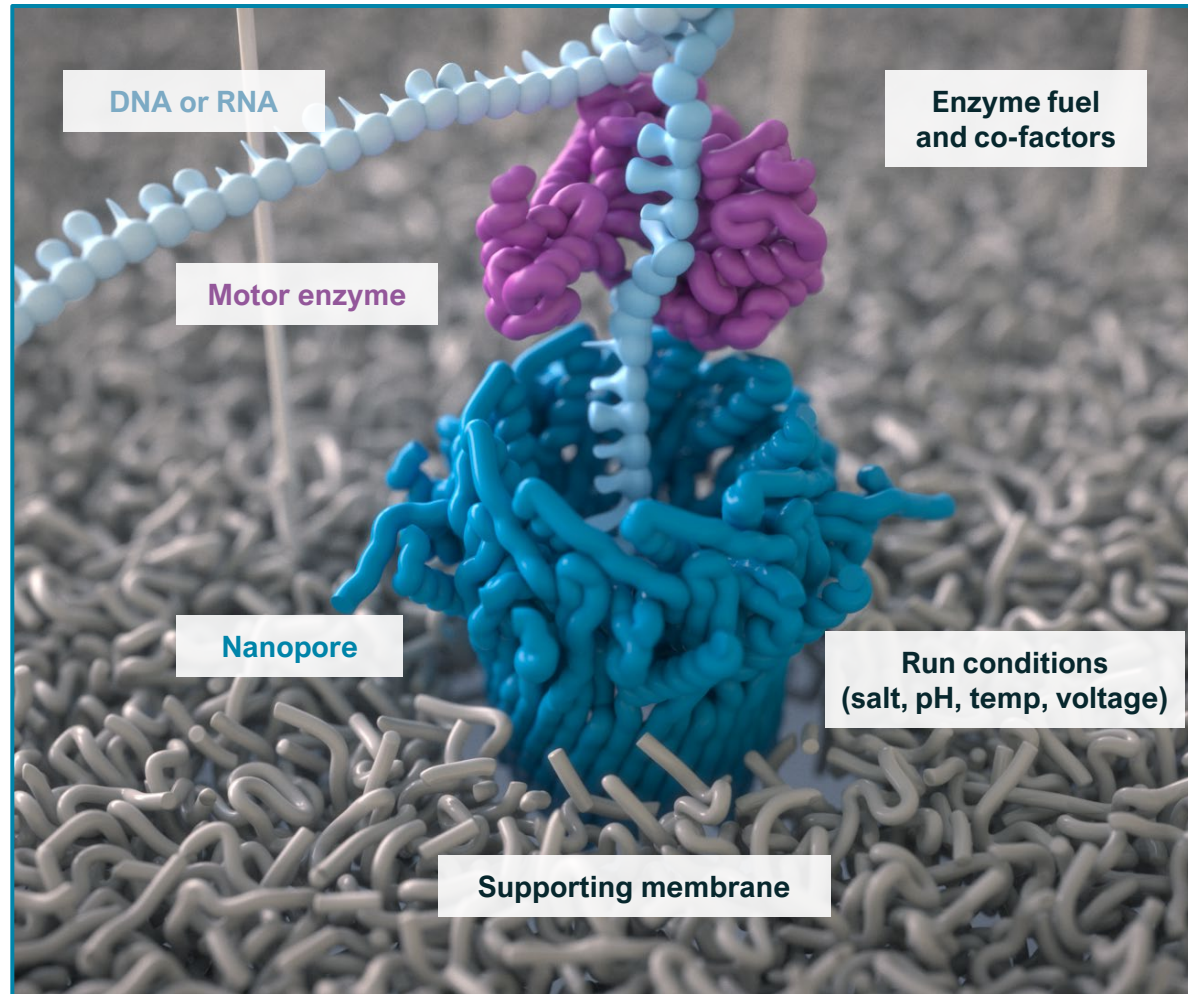


Accessible with low cost devices
and competitive price per sample

Sequence Native DNA/ RNA of any length
Adding proteins, metabolites, sugars

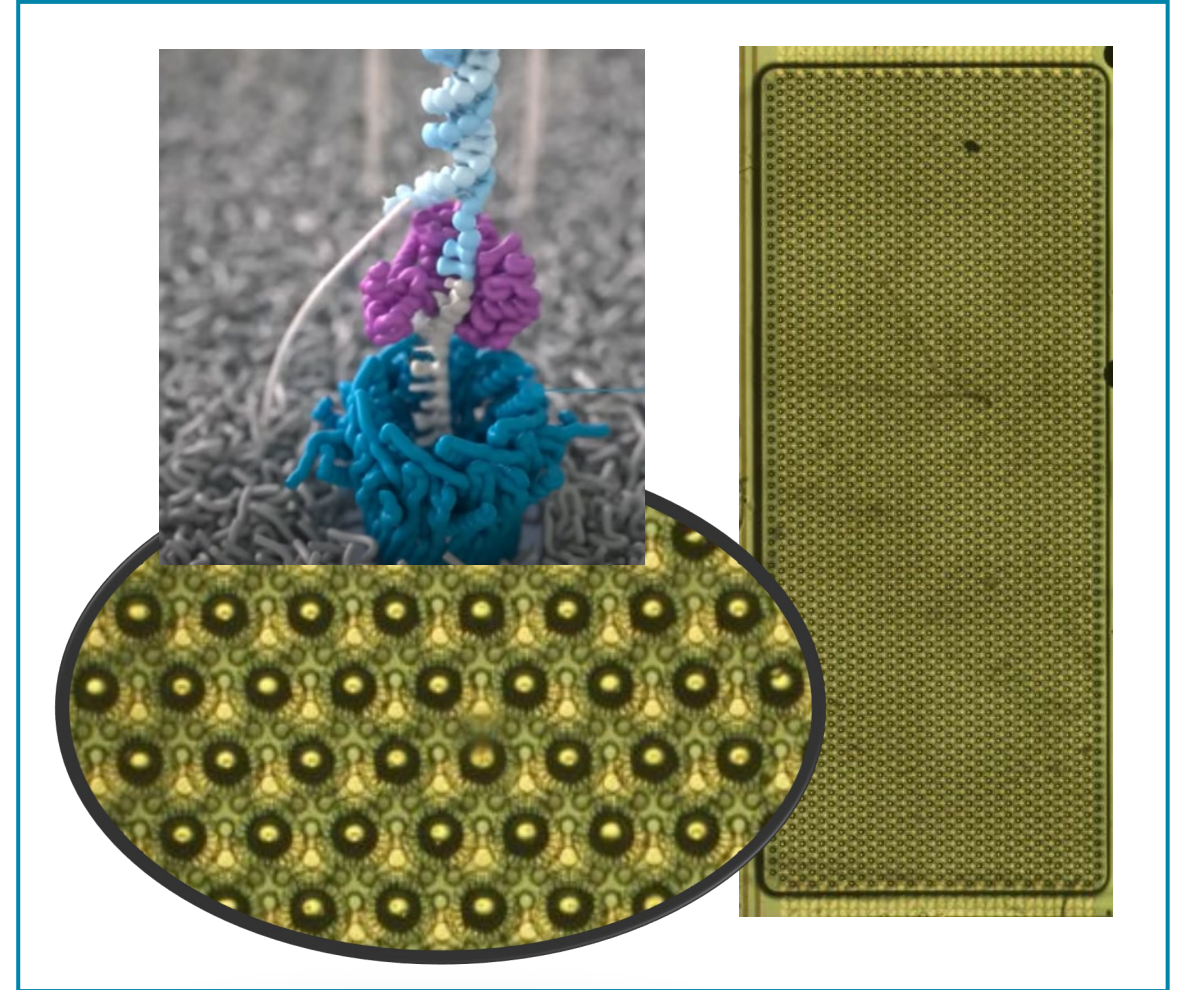
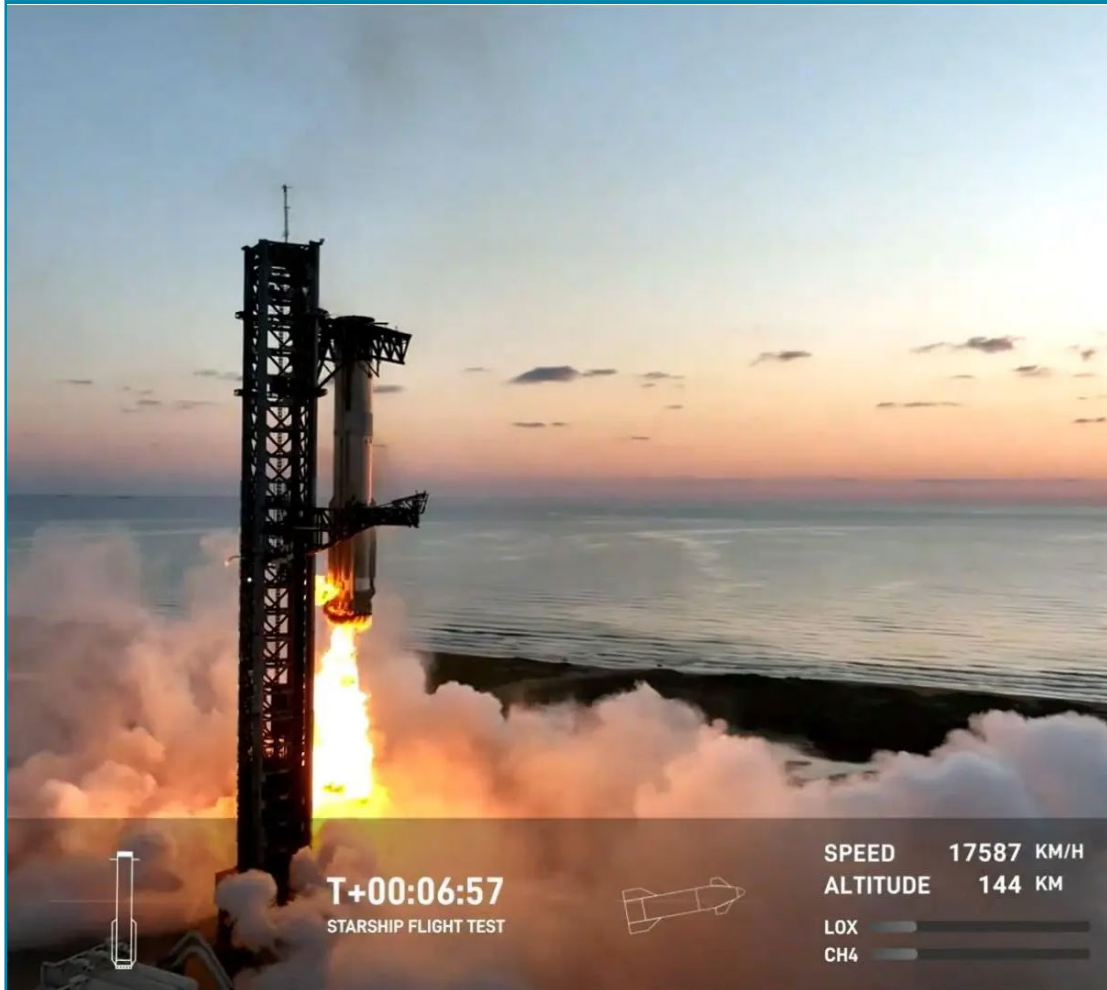
Our technology

Direct, fast, single-molecule, flexible and scalable, from small personal to high-throughput electronic devices



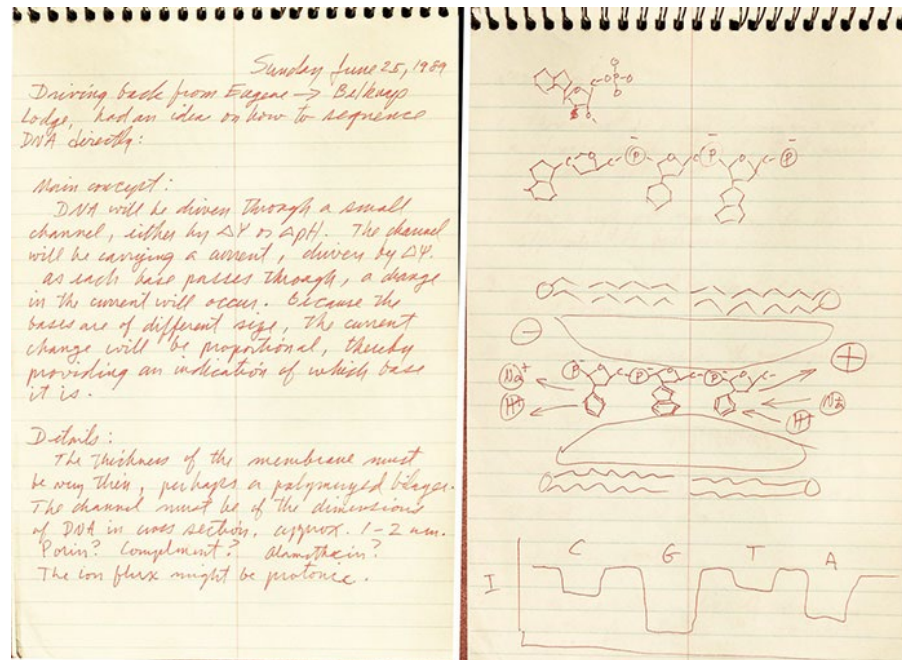
Comparable feats?

Catching rockets from space vs Catching DNA in a nanopore



The Origins of Sequencing using a Nanopore - 1989

The Golden Goose Awards 2023 – 27th September, Washington DC



Side of a Road, Oregon, 1989



Alpha Hemolysin – the ‘Original Nanopore’

Hagan Bayley’s early sensing experimentation funded by US Navy

THE JOURNAL OF BIOLOGICAL CHEMISTRY

Vol. 267, No. 25, 1992, Issue of May 23, pp. 10992-10995, 1992
Printed in U.S.A.

Functional Expression of the α -Hemolysin of *Staphylococcus aureus* in *Escherichia coli* and in Cell Lysates

DELETION OF FIVE C-TERMINAL AMINO ACIDS SELECTIVELY IMPAIRS HEMOLYTIC ACTIVITY

September 27, 1991, _____

(Received for publication, December 1, 1991)

Hagan Bayley†‡

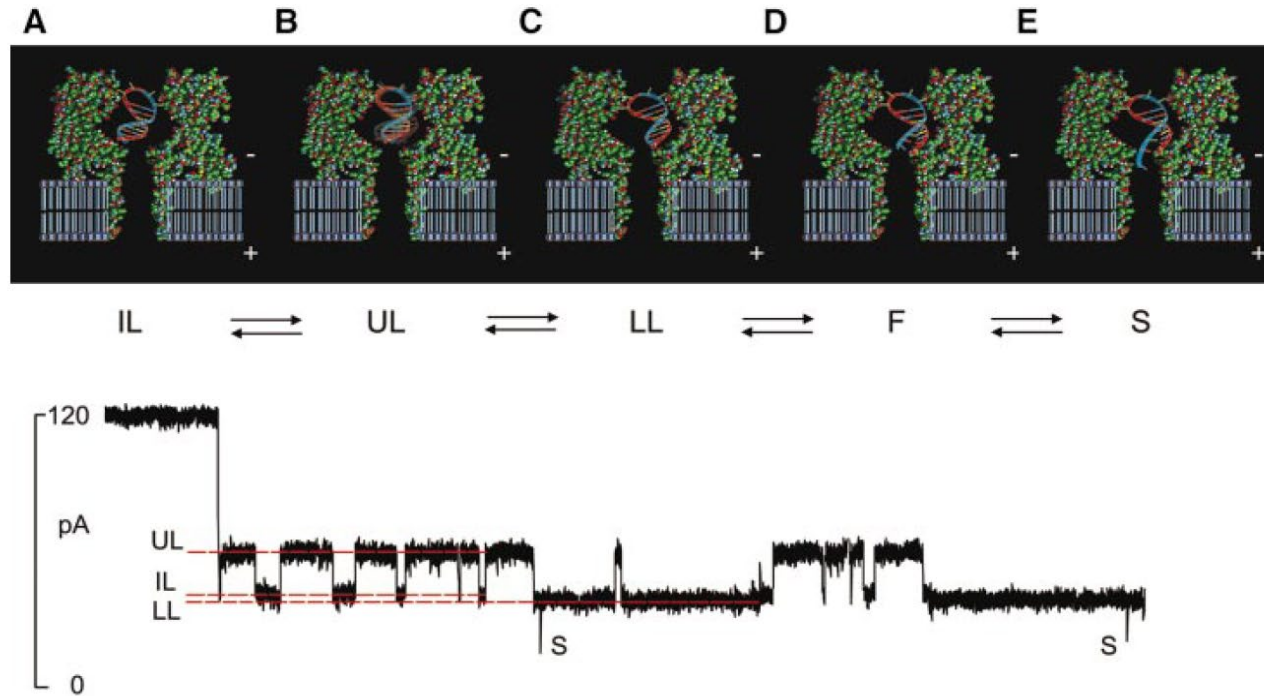
§Graduate School of
and the ¶National Institutes
National Institutes of Health,

Barbara Walker†, Musti Krishnasastri, Lynda Zorn, John Kasianowicz, and Hagan Bayley

From the ‡Worcester Foundation for Experimental Biology, Shrewsbury, Massachusetts 01545, the
Biomedical Sciences, University of Massachusetts Medical Center, Worcester, Massachusetts 01655,
of Diabetes and Digestive and Kidney Diseases-Laboratory of Biochemistry and Metabolism,
Bethesda, Maryland 20892

Early DNA base discrimination - 2003

Classifying complex features in the signal



“Discrimination among individual Watson±Crick base pairs at the termini of single DNA hairpin molecules”

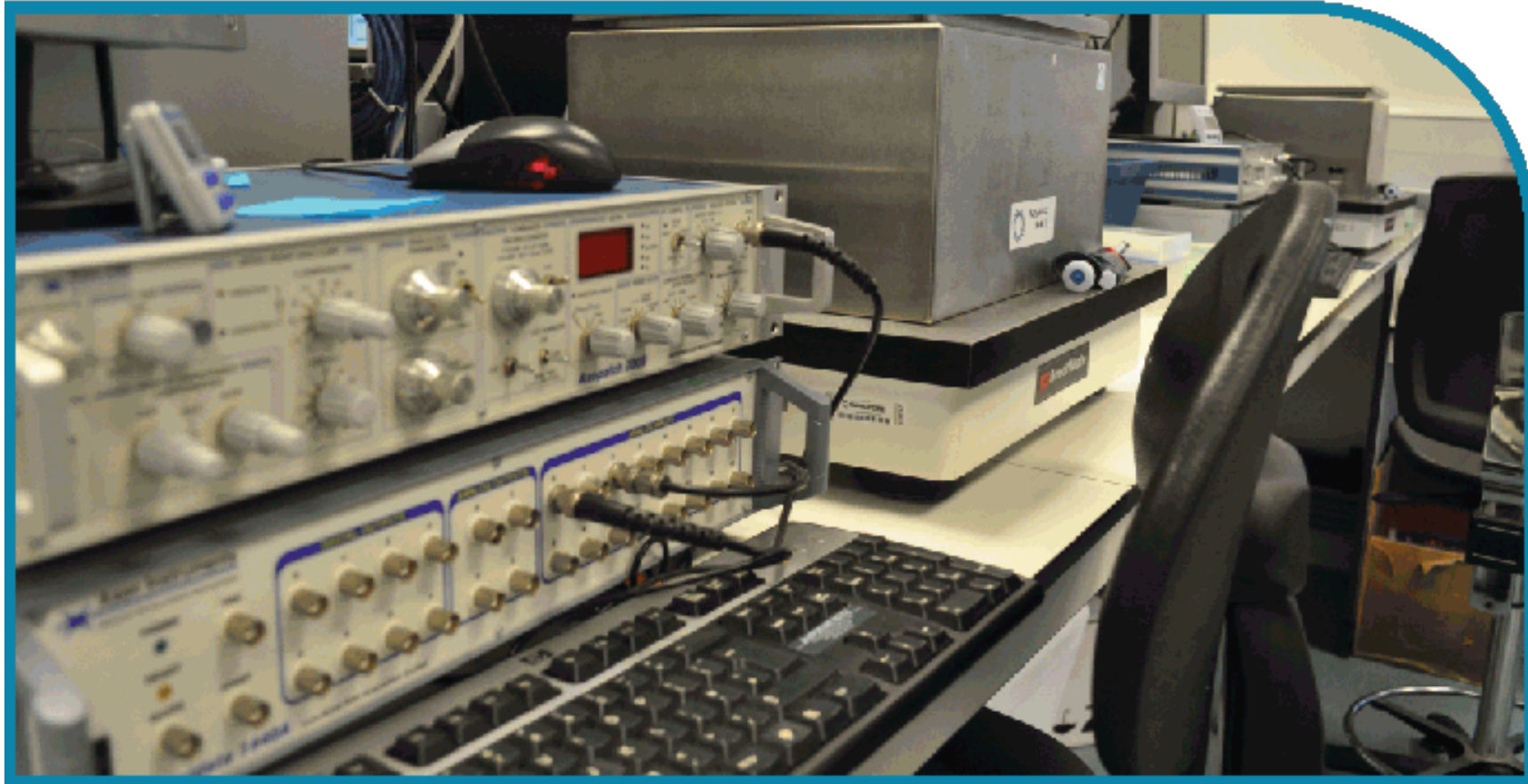
Vercoutere et al (Akeson), February 2003



L-R: Dave Deamer, Hugh Olsen, Wenona Vercoutere, Mark Akeson

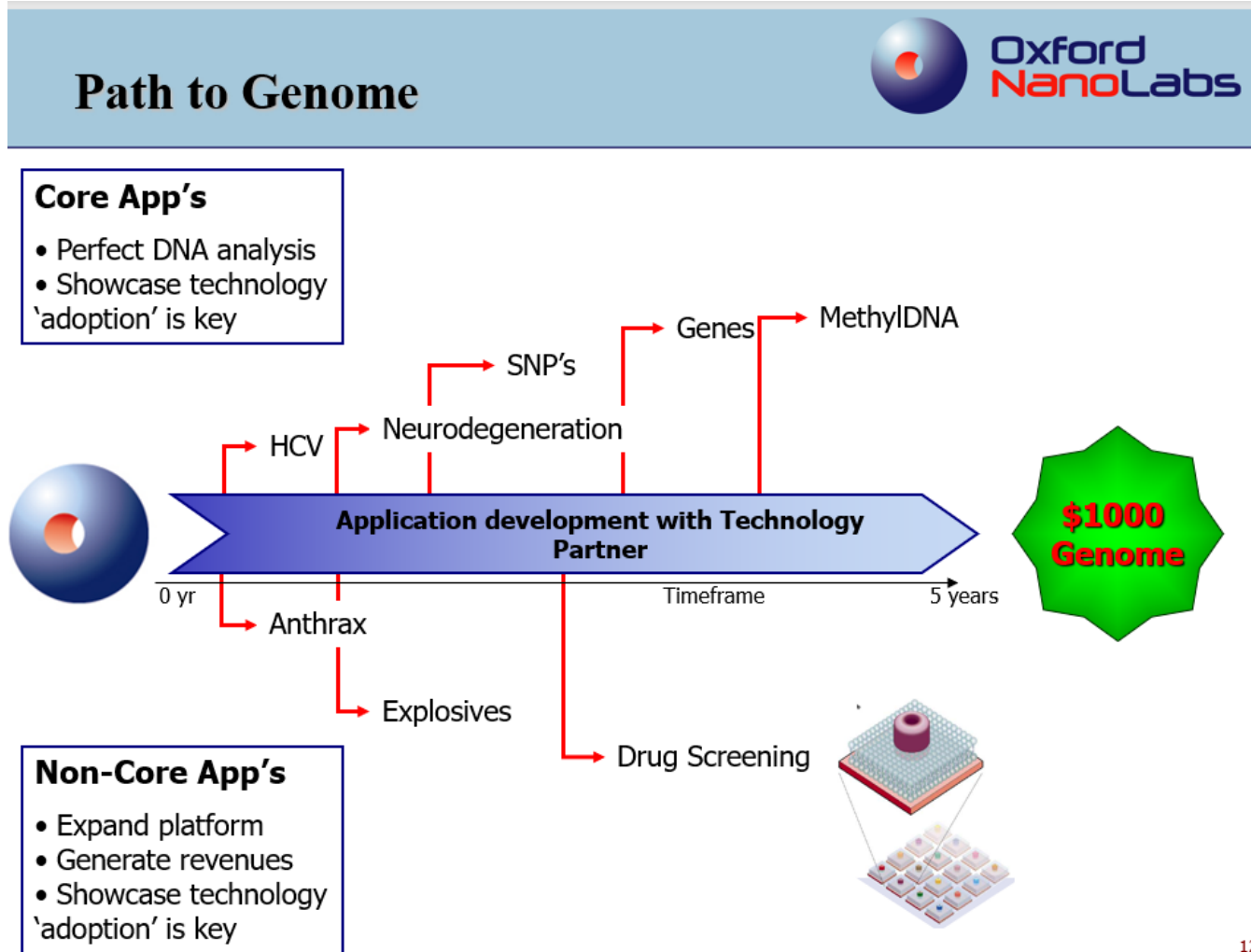
Building a Product

The Chip would be key



The Company Business Plan

From 2006 to 2007 – a change to focus on sequencing

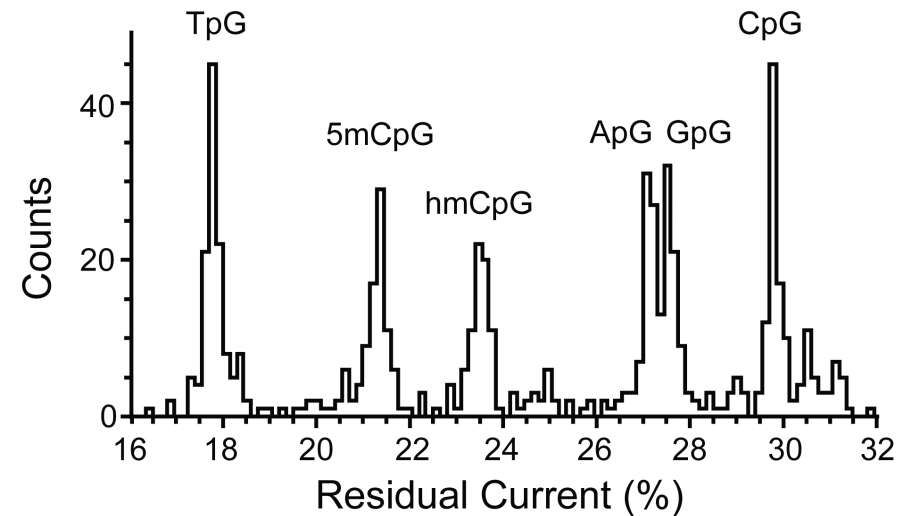
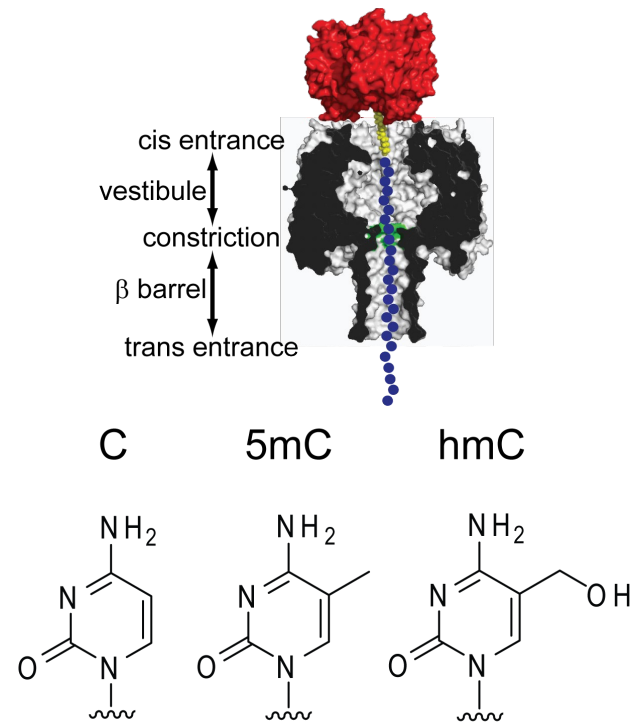


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Measurement of Static DNA Strands

Using a further mutation in aHL (NNY)

Analysis of XpG in a background of dC's



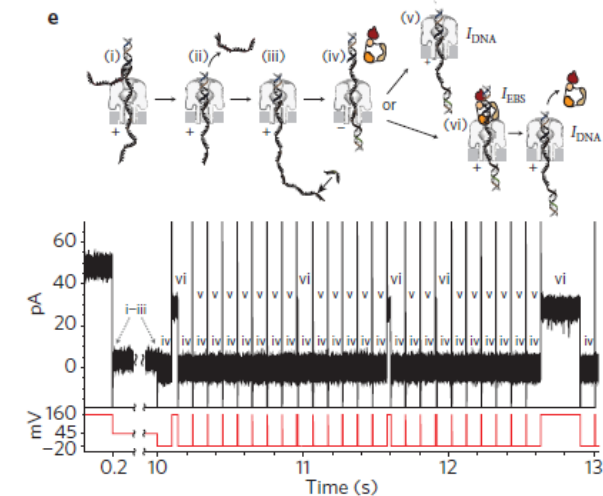
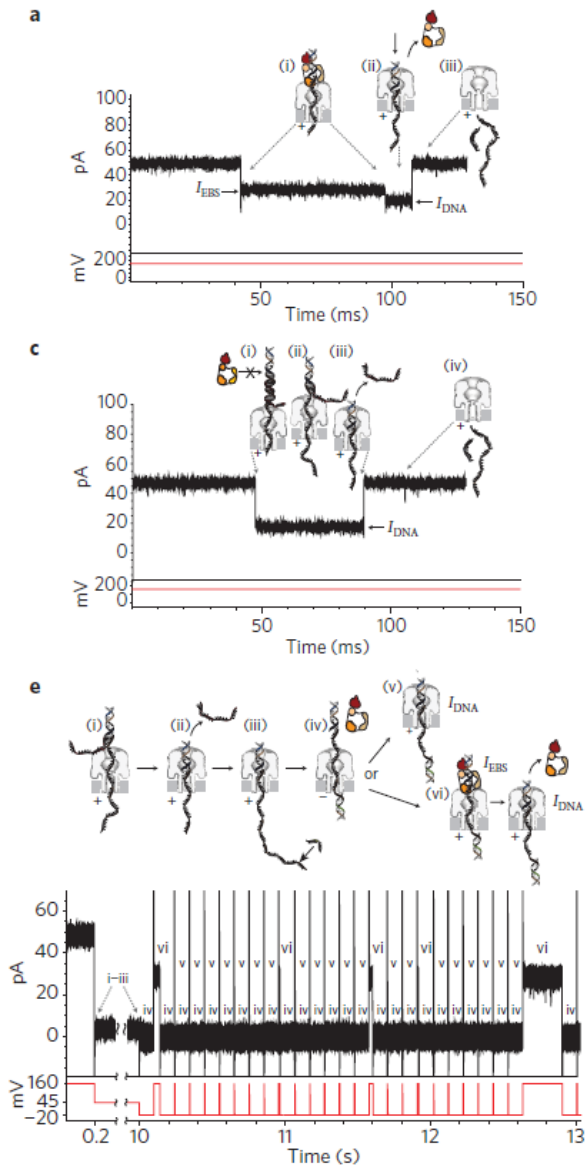
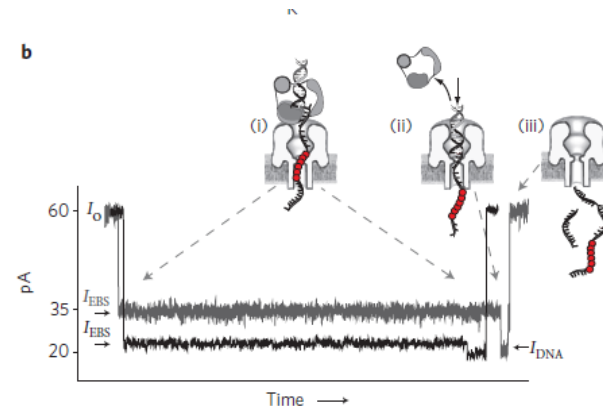
5' -C₂₀CCCCCCCCCCCCXGCCCCCCCCBt_n-3'

	I _O = 114 ± 1 pA					
Oligo	TpG	5mCpG	hmCpG	ApG	GpG	CpG
I _{RES} (%)	17.8 ± 0.1	21.3 ± 0.1	23.6 ± 0.2	27.1 ± 0.1	27.6 ± 0.1	29.8 ± 0.1

Replication of individual DNA molecules under electronic control using a protein nanopore

Felix Olasagasti, Kate R. Lieberman, Seico Benner, Gerald M. Cherf, Joseph M. Dahl, David W. Deamer and Mark Akeson*

Nanopores can be used to analyse DNA by monitoring ion currents as individual strands are captured and driven through the pore in single file by an applied voltage. Here, we show that serial replication of individual DNA templates can be achieved by DNA polymerases held at the α -haemolysin nanopore orifice. Replication is blocked in the bulk phase, and is initiated only after the DNA is captured by the nanopore. We used this method, in concert with active voltage control, to observe DNA replication catalysed by bacteriophage T7 DNA polymerase (T7DNAP) and by the Klenow fragment of DNA polymerase I (KF). T7DNAP advanced on a DNA template against an 80-mV load applied across the nanopore, and single nucleotide additions were measured on the millisecond timescale for hundreds of individual DNA molecules in series. Replication by KF was not observed when this enzyme was held on top of the nanopore orifice at an applied potential of 80 mV. Sequential nucleotide additions by KF were observed upon applying controlled voltage reversals.



Strand Sequencing – Polymerase Mode

Polymerase Mode – Real sequence (PhiX fragment)

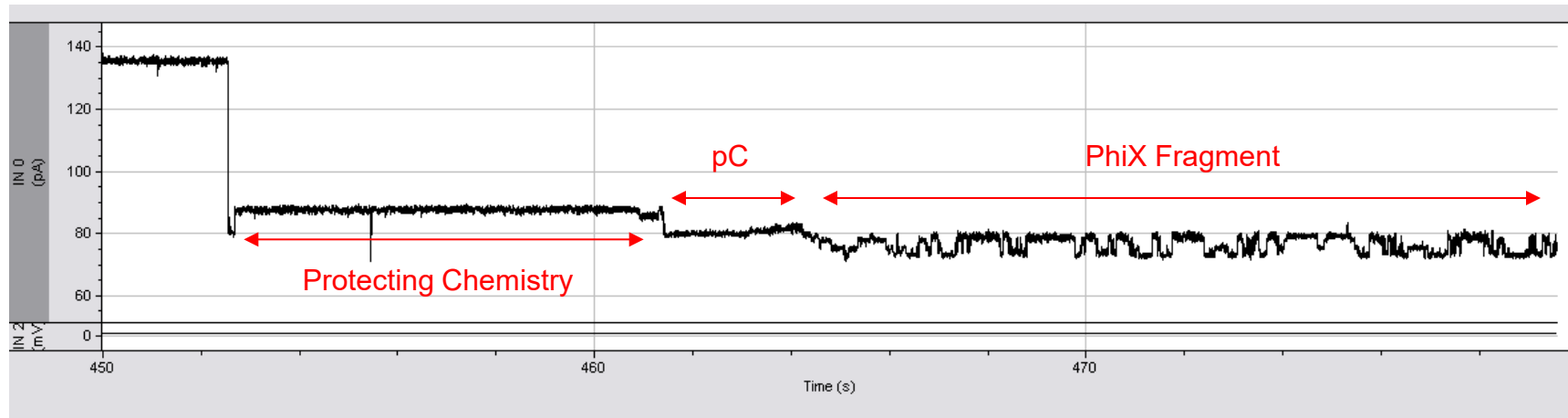
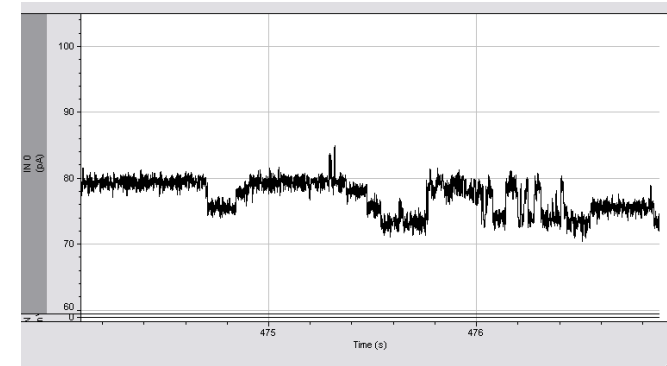
Looking at complex sequence with simple pore mutants

PolyC used as a lead-in to see sequence enter the pore

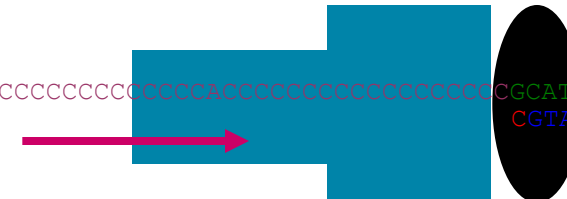
Multiple word levels seen as sequence is drawn through the pore

Improvements to base calling underway

Improve base discriminating pores under construction



CCCCCCCCCCCCCCCCGAGTTTTATCGCTTCCATGACGCAGAAGTTAACACTTTCGGATATTTCTGATGAGTCGAAAAATTATCTTCCCCCCCCCCCCCCCCCGCATGCGAGGTAGCCTT
CGTACGTCCATCGGTT



Defining Features of Nanopore-based Sequencing vs SBS

Sequencing by Synthesis

- Native molecule is converted to a synthetic form.
- Samples are fixed to sensor surface.
- Each base is measured in isolation with confidence value.
- Sequencing 'run' ends when the fragment is read.
- Data can only be processed at the end of the run.
- Monolithic sensor design not modular.
- Large sensor reduces cost per read.

Run time = Read Length = Output = Quality = Cost/Base
Price of Instrument = Price of Run = Cost/Base

Nanopore-based Sequencing

- Native molecule is measured at 10-100,000 faster than SBS.
- Read length = Fragment Length
- Samples are free flowing – wash/reload
- Each base is measured many times in combination with other bases.
- Analysis can occur during the run
- Sequencing 'run' time per sample is variable.
- Time to sequence a fragment << experiment time
- Modular design means costs scale linearly

Independence of Run time, Read length, Output, Cost

The O.G. (Original GridION)

Scalability, Versatility, Efficiency using
Distributed, Real-time analysis with the GridION System



Scalable, groupable

Desktop



Mid range installation



Supercenter



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CONFIDENTIAL

 **NANOPORE**
Technologies

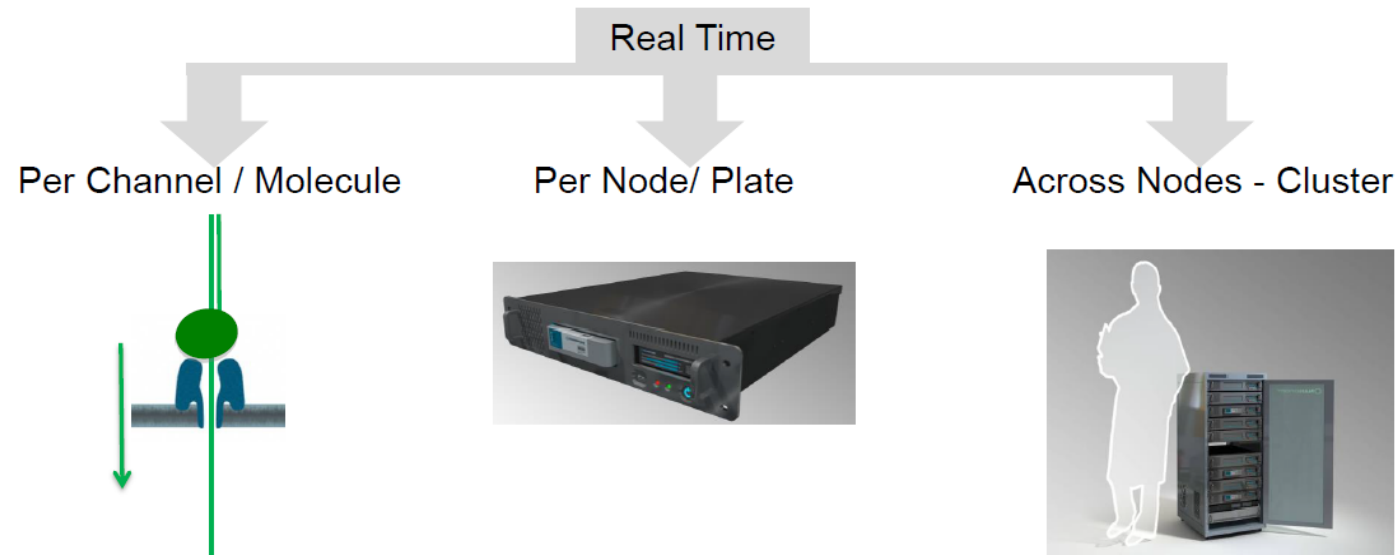
8

The O.G.

Key feature

Run Until ... Sufficient

- ⚙ Data Quantity
- ⚙ Data Quality
- ⚙ Specific Observations Made (for application)



© 2011 Oxford Nanopore Technologies Ltd

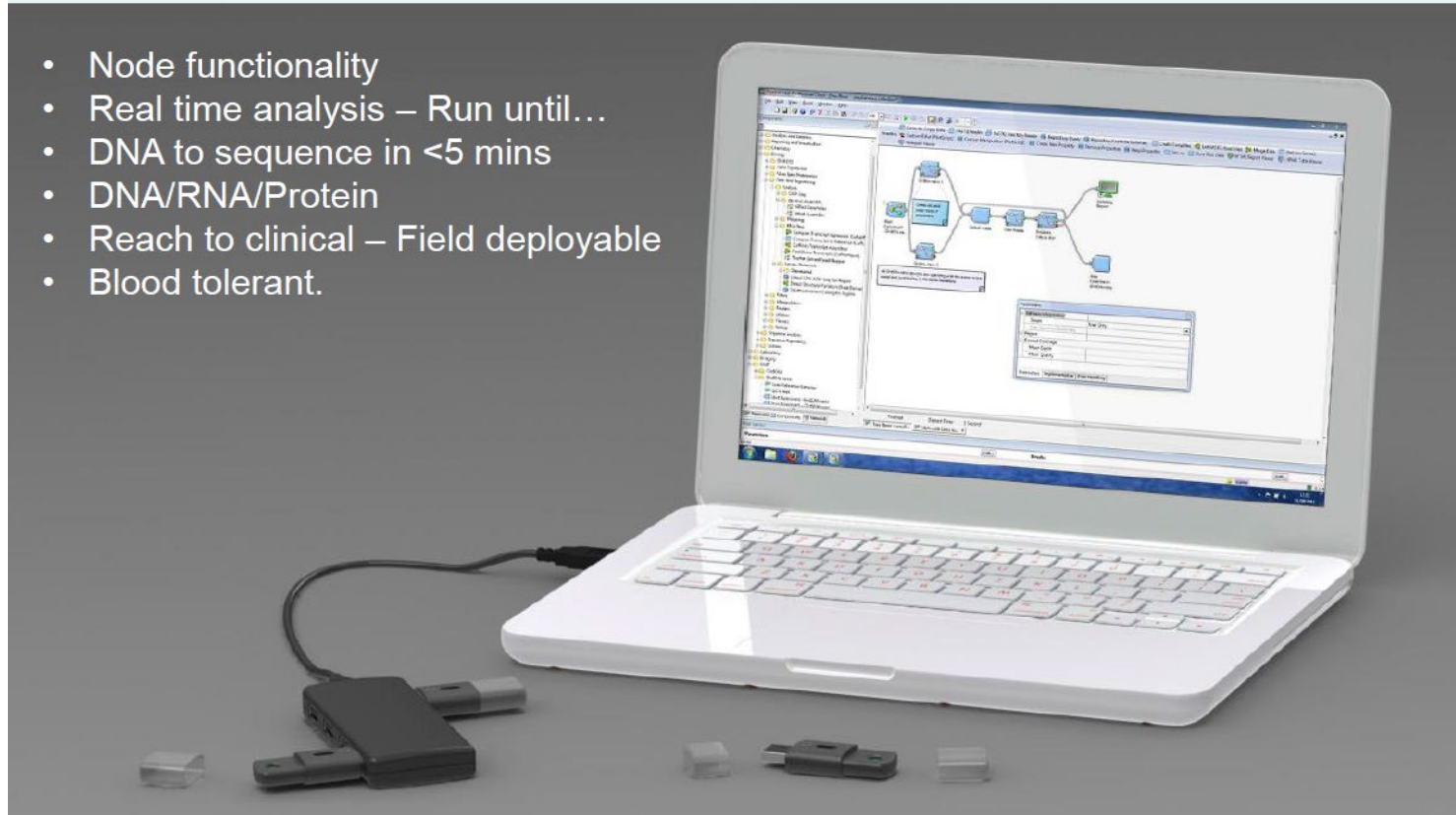
CONFIDENTIAL

The O.G.

Factory-made

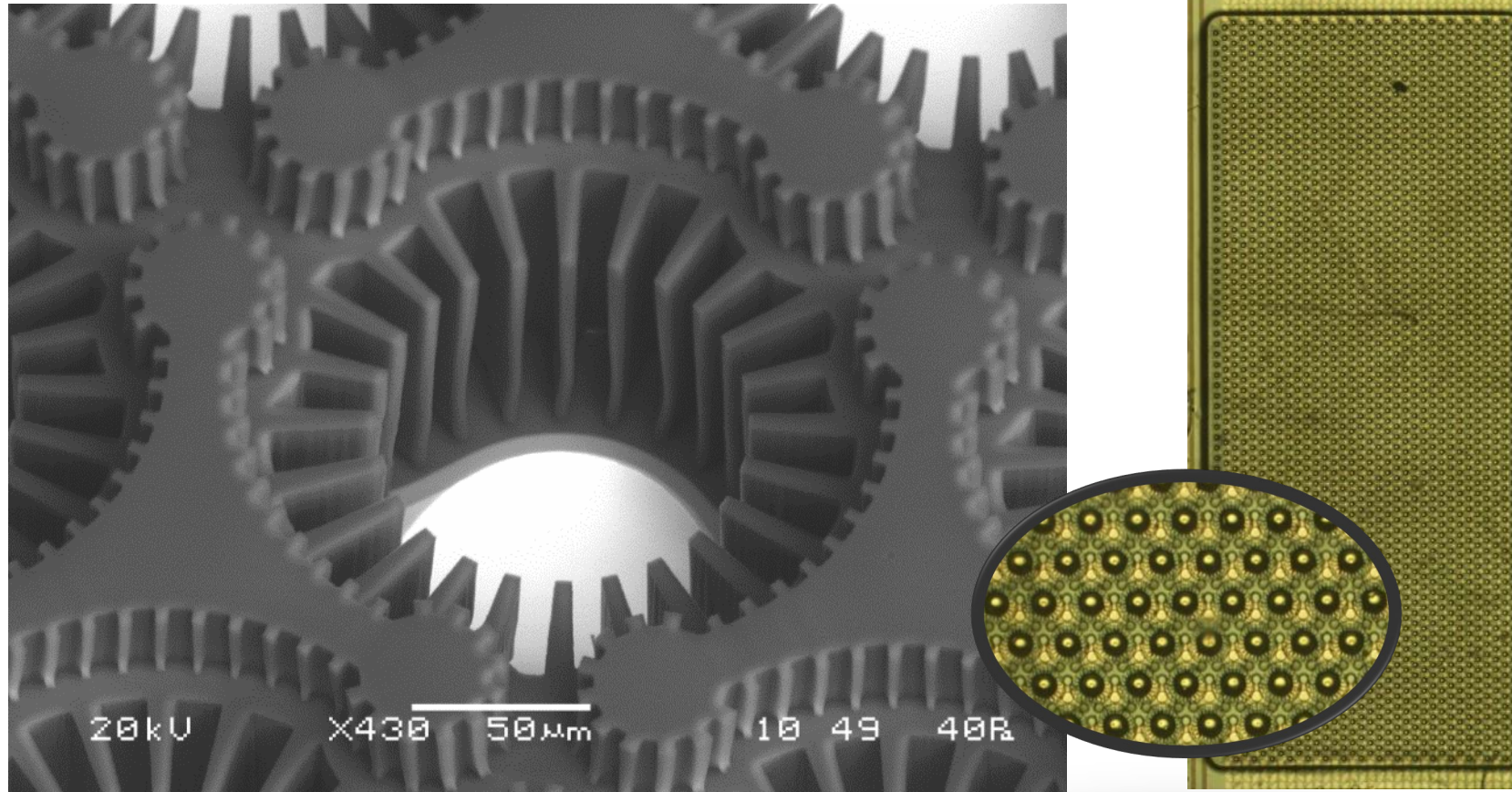
“GridION Nano”

- Node functionality
- Real time analysis – Run until...
- DNA to sequence in <5 mins
- DNA/RNA/Protein
- Reach to clinical – Field deployable
- Blood tolerant.



Self-assembling, self-stabilising system

Inside the MinION flow cell



First published Nanopore read, 11th June 2014

Courtesy of Nick Loman, Josh Quick, Szymon Calus

A *P. aeruginosa* serotype-defining single read from our first Oxford Nanopore run

Cite

Download (0 kB)

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Embed

+ Collect

Dataset posted on 11.06.2014, 07:38 authored by **Nicholas Loman**, Josh Quick, Szymon Calus

Here is, I think, the first publically-available Oxford Nanopore read to be published. This came off our MinION instrument this morning (Wednesday 11th June). The DNA was derived from *Pseudomonas aeruginosa* strain 910 which originally came from hospital water. DNA was fragmented with Covaris G-Tube as per the Oxford Nanopore standard genomic library preparation protocol. The read maps to part of the *P. aeruginosa* O6-antigen determining region.

HISTORY

11.06.2014 - First online date, Posted date

USAGE METRICS

9800
views

451
downloads

2
citations



Nick Loman @pathogenomenick@mstdn.science
@pathogenomenick

A genuine >8kb Oxford @nanopore read off our instrument THIS MORNING. And happily it's actually useful for diagnosis! figshare.com/articles/A_P_a...

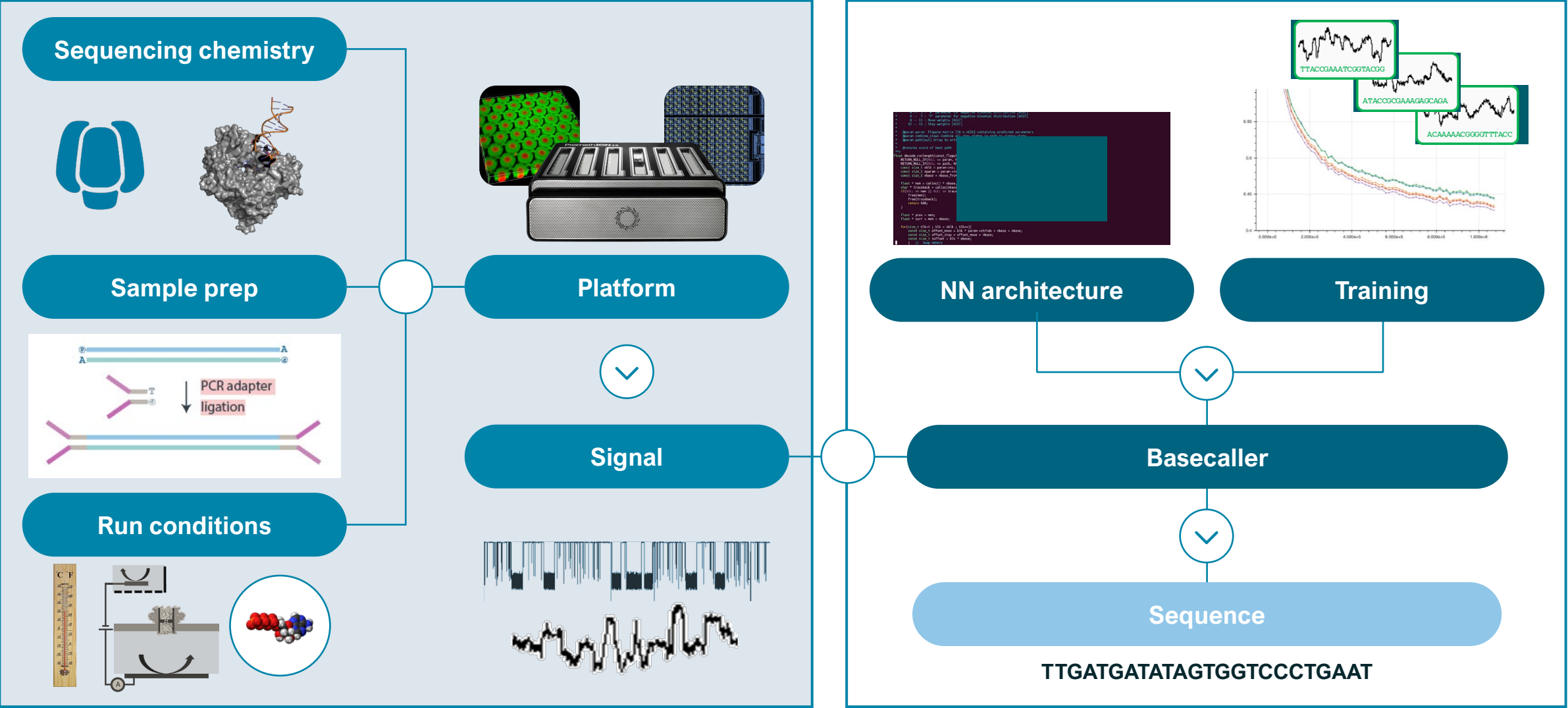
7:40 AM · Jun 11, 2014 · Tweetbot for Mac

108 Retweets 2 Quote Tweets 40 Likes



https://figshare.com/articles/dataset/A_P_aeruginosa_serotype_defining_single_read_from_our_first_Oxford_Nanopore_run/1052996

Pushing our platform performance

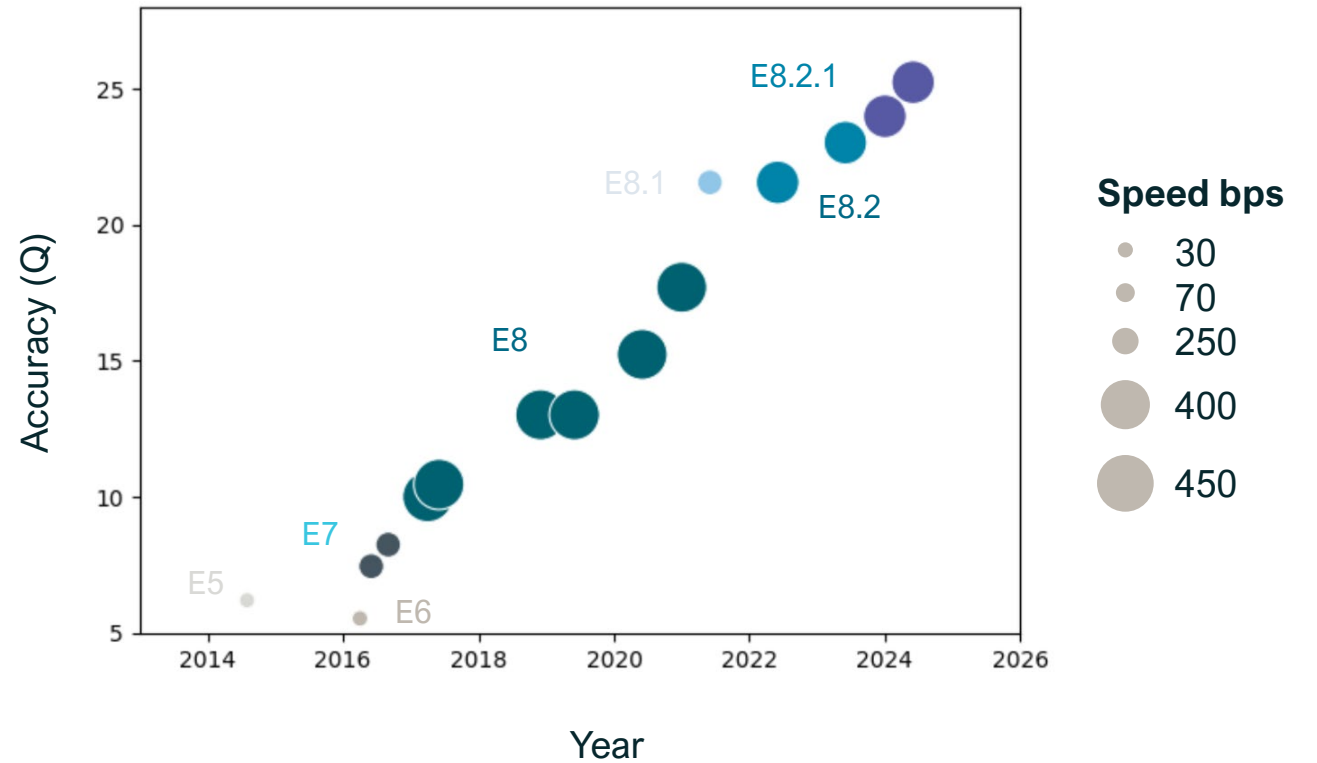


Evolution of motors

Enzyme engineering

- Motor controls the movement of **DNA / RNA** through the pore
- Motor performance **increased** over the years
 - ✓ Faster motors increase output
 - ✓ Movement consistency improves accuracy
 - ✓ Motor processivity increased read length
 - ✓ Motor stability extends run time
- AlphaFold-like **guided designs**
- **Automated** screening pipelines in place

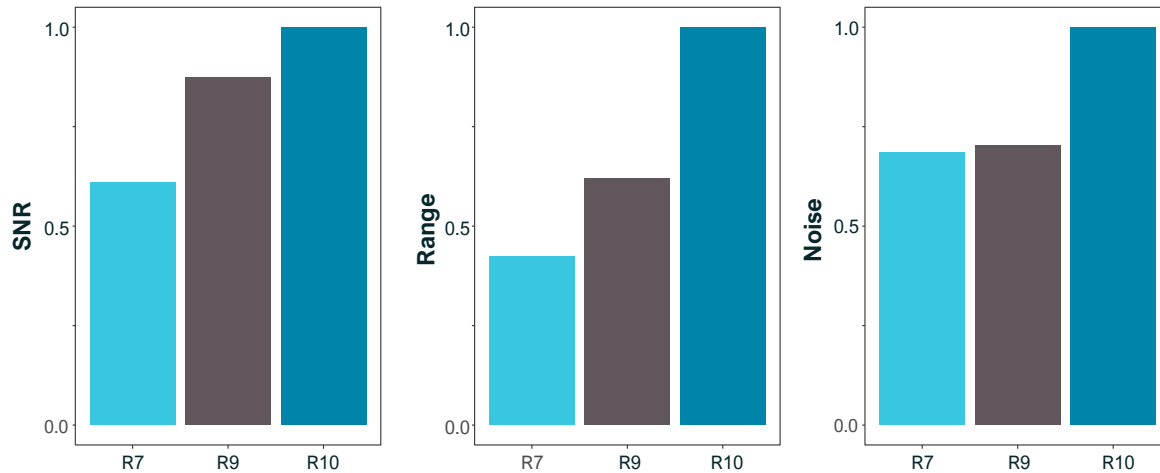
Motor upgrades



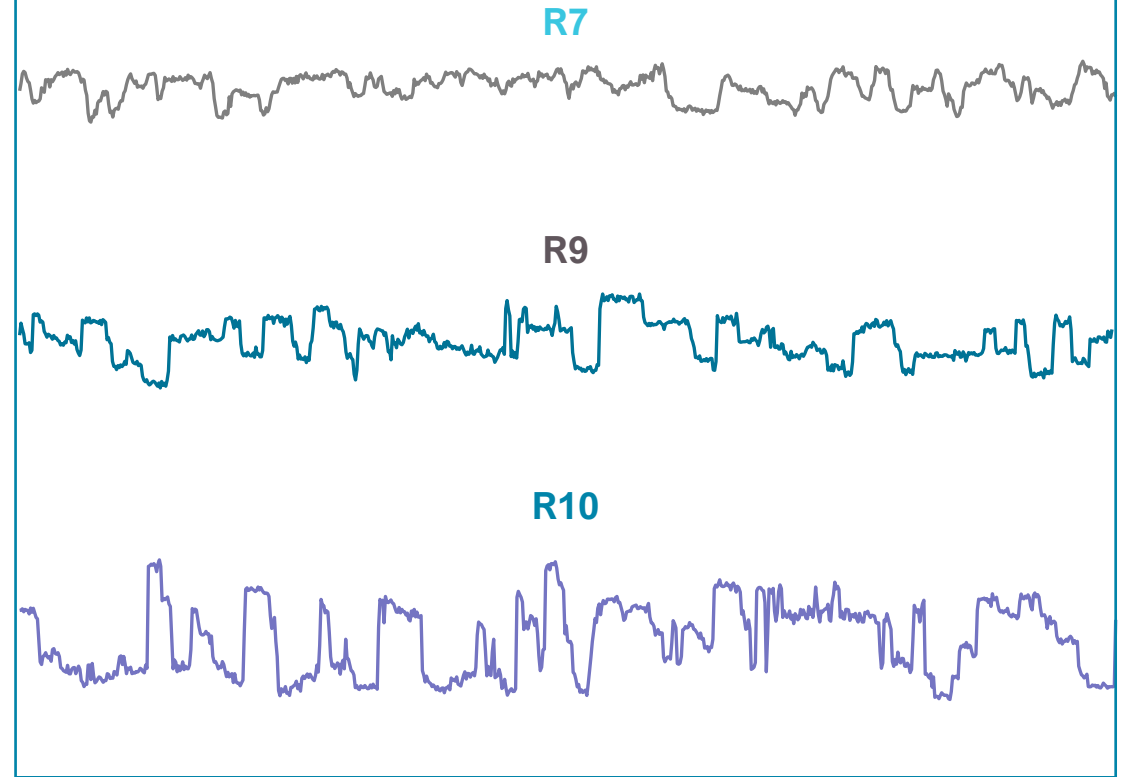
Evolution of nanopores

Designed to improve base resolution and resolve low complexity signal

- Significant accuracy gains have come from **upgrading** the class of reader
- Minor revisions have increased **stability, robustness** and **sensitivity**
- Over **11,000 pore mutants** have been tested in-house to date



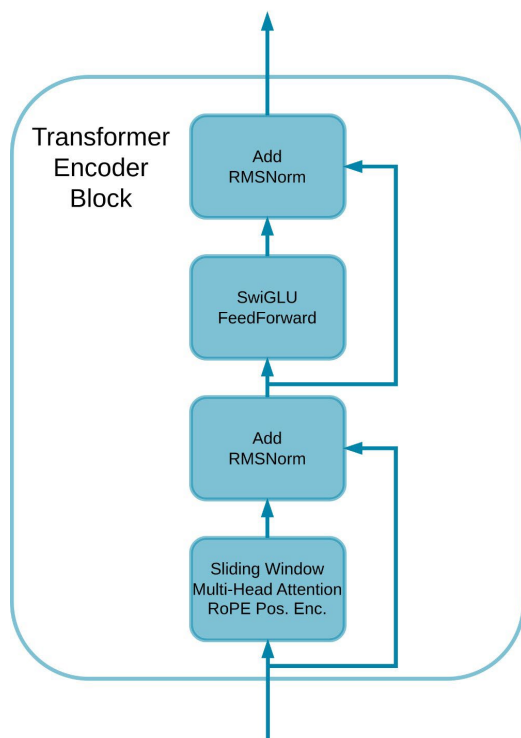
Pore upgrades



Transformer models (V5 SUP)

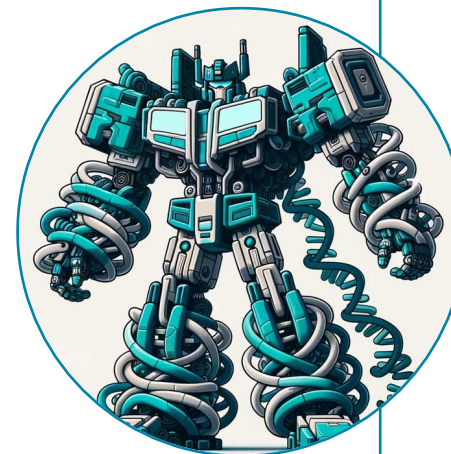
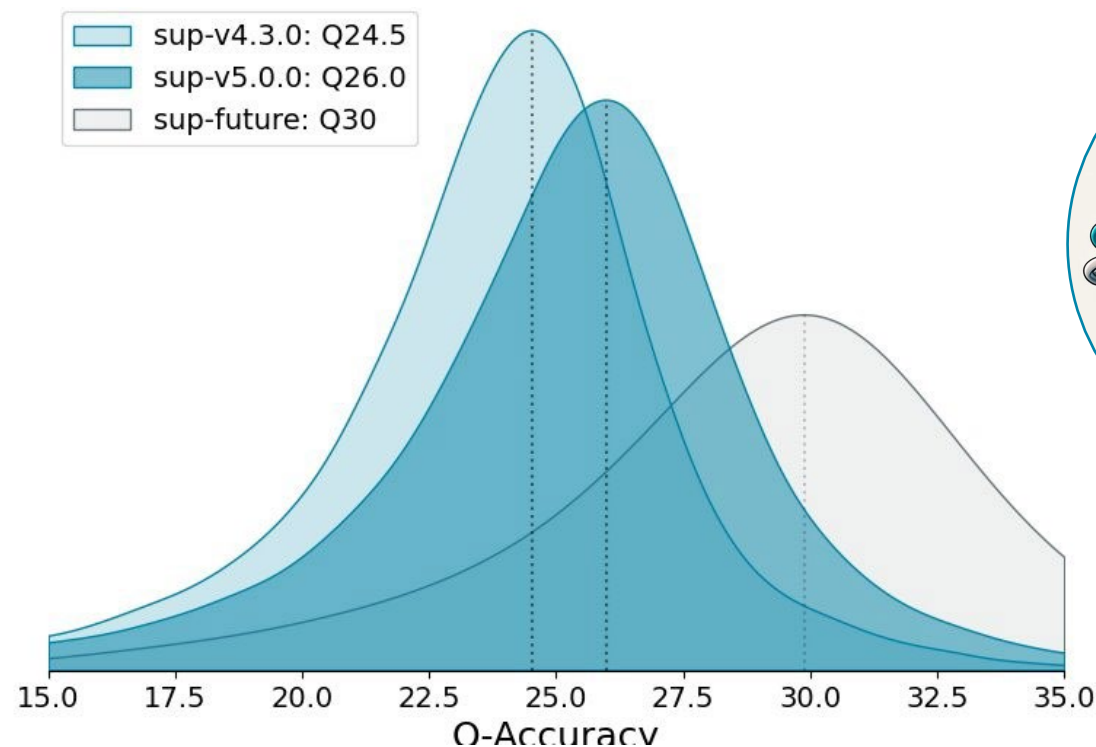
Oxford Nanopore basecalling model architecture updated in 2024

- Model architecture inspired by **Llama 3**
- HAC models remain **LSTM**



Now in Dorado 0.7

Latest accuracy

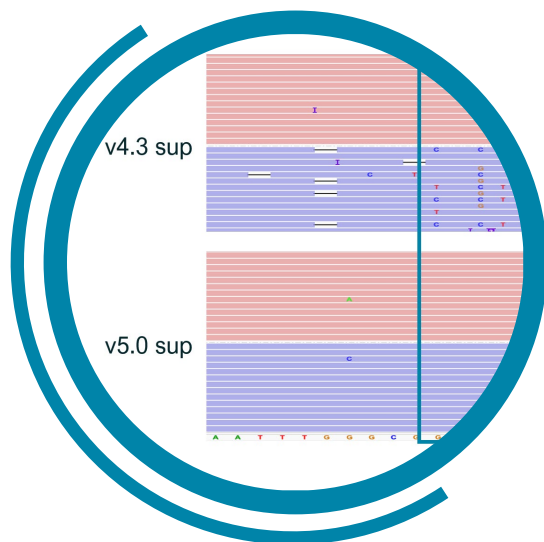


* Modal accuracy measured on 10x HG002:chr6_MATERNAL holdout dataset of r10.4.1-e8.2.1 data using Dorado-0.7 with the v4.3.0-sup + v5.0.0-sup basecall model

Summary of ML areas of use

Employing ML improvements to basecalling and beyond

Basecalling algorithms

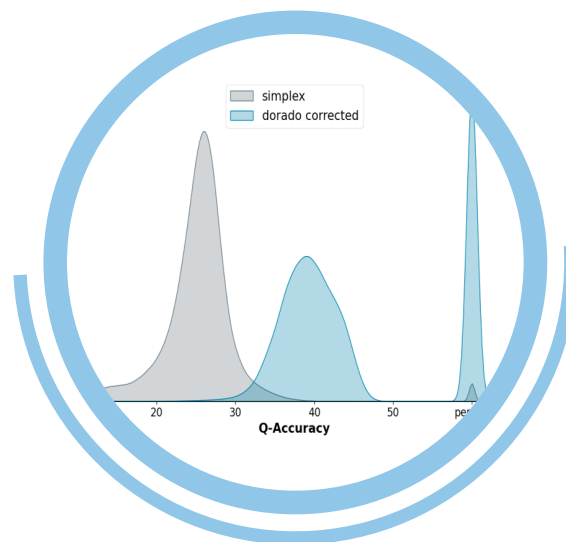


Improving basecaller training sets to include diverse modifications in diverse contexts

Increases consensus accuracy

Latest improvement delivers >Q50 bacterial assemblies on a single technology

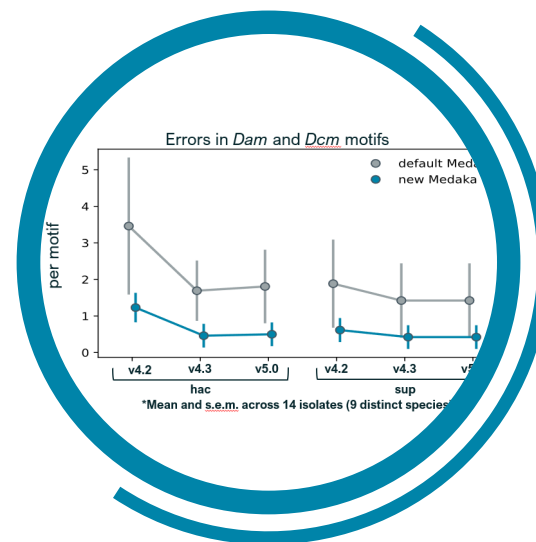
Pre-assembly read correction



All-vs-all alignment of reads followed by haplotype-aware correction achieves extremely high single-read accuracies approaching Q40

Collaboration with Mile Šikić AI in Genomics Lab at the Genome Institute of Singapore to develop HERRO

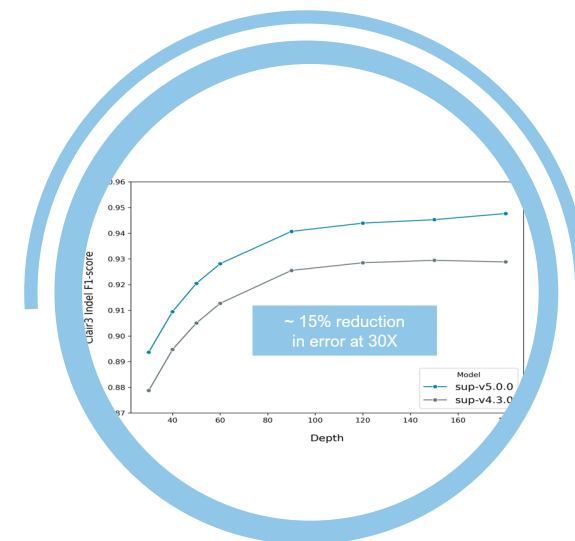
Consensus polishing



New Medaka model provides complementary improvements in combination with recent basecaller versions

Highly effective for plasmid sequencing: robust improvements in common *E. coli* motifs, using either hac or sup

Variant calling



Improvements to Clair3 training processes give a big improvement to indel calling at inference time

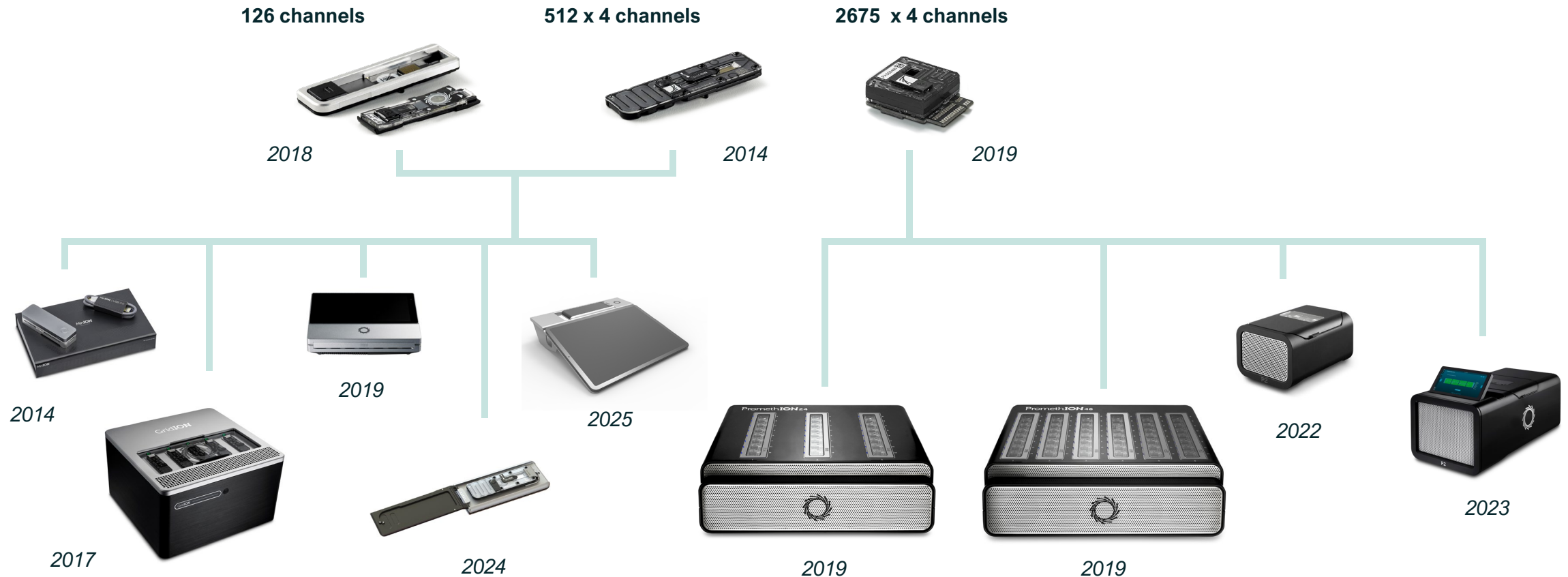
SNP accuracy maintained at 99.9%

Further improvements targeted at reducing difference in variant calling performance based on raw read accuracy.

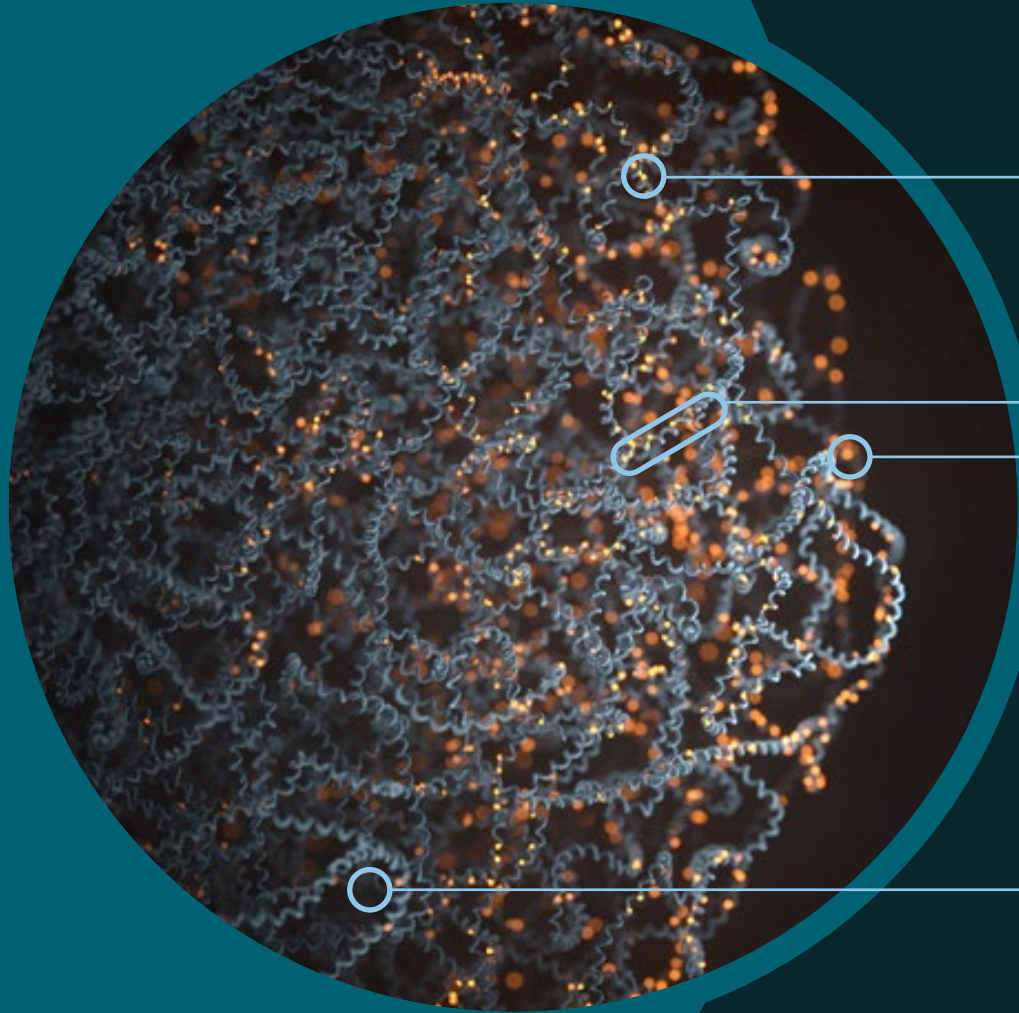
Collaboration with Ruibang Luo, HKU

Nanopore Platform

Product family: Flongle + MinION + PromethION



Finally uncover all areas of the human genome



Small variants

SNPs and INDELs well characterised by existing technology. E.g. Sickle cell disease and cystic fibrosis

Larger, more complex variants

Structural variants including Copy Number Variants, Repeats, Expansions only fully visible with the advent of long reads. E.g. a Alzheimer's , parkinsons

Epigenetics

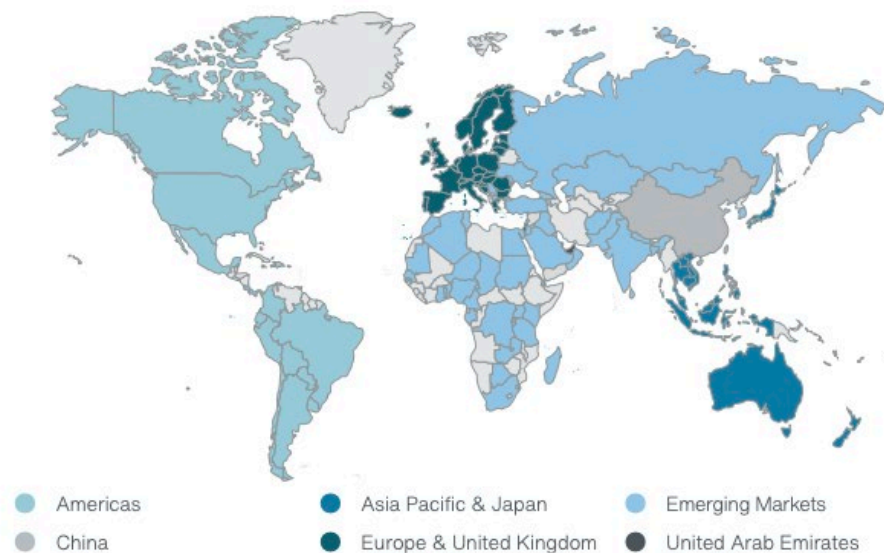
Methylation, highly involved in complex disease e.g. cancer, is only partially visible through modified chemistries or algorithmic interpretations is now unlocked with native nanopore sensing

Dark genome

Medically important areas of the genome that have only recently been uncovered and only visible through native DNA or RNA sequencing

Since the MinION Launch in 2014

An ever expanding community across the planet



> 10¹⁹ nanopores

>120

Countries

>40,000

Community members

>14,000

Papers published by the Nanopore Community

Cassava virus sequencing for farmers

Laura Boykin Okalebo, Joseph Ndunguru and the Cassava Virus Action Project (2017)



**Food Security: Using genomics
to fight cassava pathogens**

Sequencing by young(er) Scientists

What's in My Sandwich by Sophie Zaajier, #DaffDNA by Jon Hale, Pet Angelfish genome by Indeever Madireddy

Cutting Edge: Using mobile sequencers in an academic classroom

Sophie Zaajier, Columbia University Ubiquitous Genomics 2015 class, Yaniv Erlich

Columbia University, United States; New York Genome Center, United States; Center for Computational Biology and Bioinformatics, Columbia University, United States

Apr 7, 2016 · <https://doi.org/10.7554/eLife.14258>



← The Daffodil DNA Project
149 Tweets



... ⓧ ↻ Following

The Daffodil DNA Project
@DaffDNA Follows you

A collaborative project between schools and scientists funded by @royalsociety #partnershipgrants to decode the genetics of daffodils

Heredity (2020) 125:431–436
<https://doi.org/10.1038/s41437-020-00370-0>

the **genetics**SOCIETY

PERSPECTIVE

Engaging the next generation of plant geneticists through sustained research: an overview of a post-16 project

Jon Michael Hale

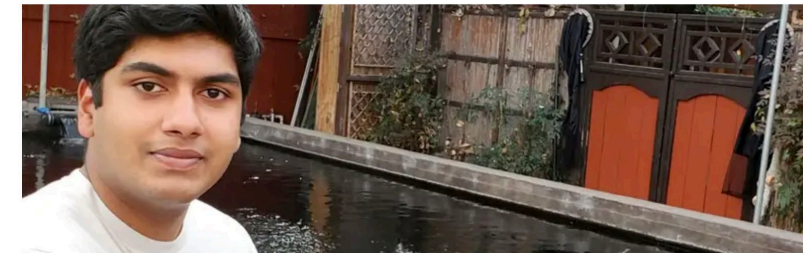
High school student is first to sequence the angelfish genome

17-year-old Indeever Madireddy sequenced the genome of his pet angelfish after it died – the first time this species has been sequenced



LIFE 21 October 2022

By Michael Le Page



Rapid sequencing in prenatal and critical care

Steven Jones (BCGSC), Euan Ashley (Stanford), Danny Miller (Uni of Washington)



CLINICAL REPORT

An approach to rapid characterization of *DMD* copy number variants for prenatal risk assessment

Hui-Lin Chin, Kieran O'Neill, Kristal Louie, Lindsay Brown, Kamilla Schlade-Bartusiak, Patrice Eydoux, Rosemarie Rupps, Ali Farahani, Cornelius F. Boerkoel ✉, Steven J. M. Jones ... See fewer authors ^

First published: 21 May 2021 | <https://doi.org/10.1002/ajmg.a.62349>

“importantly it did allow for a timely and accurate counselling advice. So there is a baby alive today because of Oxford Nanopore technology.”

- Steven Jones, email 11 March 2021

Ultrarapid Nanopore Genome Sequencing in a Critical Care Setting

TO THE EDITOR: Rapid genetic diagnosis can guide clinical management, improve prognosis, and reduce costs in critically ill patients.^{1,2} Although most critical care decisions must be made (the average of postsequencing run times for Patients 2 to 12) (Table S5). Flow cells were washed and reused until exhaustion to reduce the sequencing cost per

John E. Gorzynski et al
NEJM, January 2022, DOI: 10.1056/NEJMc2112090

medRxiv



BMJ Yale

3-hour genome sequencing and targeted analysis to rapidly assess genetic risk

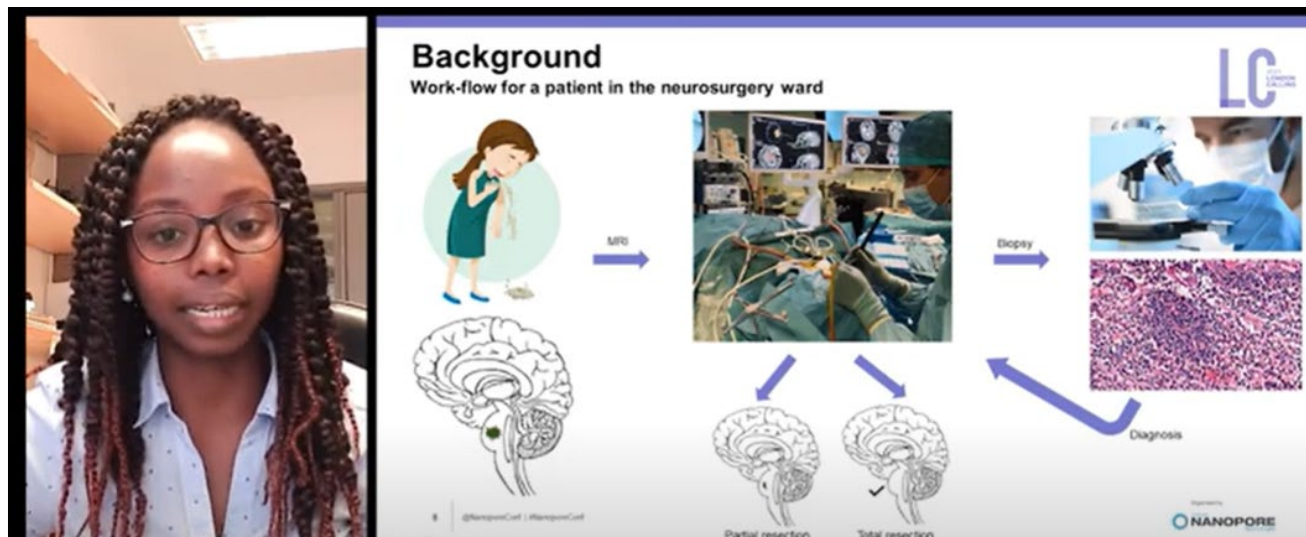
Miranda Galey, MS^{1,2}, Paxton Reed, BS^{1,2}, Tara Wenger, MD, PhD¹, Erika Beckman, MS, CGC¹, Irene J. Chang, MD¹, Cate R. Paschal, PhD^{2,3}, Jillian G. Buchan, PhD², Christina M. Lockwood, PhD^{2,4,5}, Mihai Puia-Dumitrescu, MD⁶, Daniel R. Garalde, PhD⁷, Joseph Guillory, MS⁷, Androo J. Markham, MS⁷, Andrew B. Stergachis, MD, PhD^{4,5,8}, Michael J. Bamshad, MD^{1,4,5}, Evan E. Eichler, PhD^{4,5,9}, Danny E. Miller MD, PhD^{1,2,5,✉}

¹ Division of Genetic Medicine, Department of Pediatrics, University of Washington and Seattle Children's Hospital, Seattle, WA, USA



Intraoperative Brain Tumour classification

Luna Djirackor (Oslo Uni), Matt Loose (Nottinham Uni), Christiaan Vermeulen (UMC Utrecht)



Article

Ultra-fast deep-learned CNS tumour classification during surgery

<https://doi.org/10.1038/s41586-023-06615-2>

Received: 10 February 2023

Accepted: 6 September 2023

Published online: 11 October 2023

Open access

Check for updates

C. Vermeulen^{1,2,6}, M. Pagès-Gallego^{1,2,6}, L. Kester³, M. E. G. Kranendonk³, P. Wesseling^{3,4}, N. Verburg⁵, P. de Witt Hamer⁵, E. J. Kooi⁴, L. Dankmeijer^{4,5}, J. van der Lugt³, K. van Baarsen³, E. W. Hoving³, B. B. J. Tops³ & J. de Ridder^{1,2}✉

Central nervous system tumours represent one of the most lethal cancer types, particularly among children¹. Primary treatment includes neurosurgical resection of the tumour, in which a delicate balance must be struck between maximizing the extent of resection and minimizing risk of neurological damage and comorbidity^{2,3}. However, surgeons have limited knowledge of the precise tumour type intraoperatively.

Mini-symposium: neuropathology

Nanopore-based brain tumour classification: the harbinger of near-patient, ultra-rapid tumour sequencing

Simon Deacon, Matt Loose, Stuart Smith, Simon Paine

C. Vermeulen et al

Nature, 1 October 2023, DOI: 10.1038/s41586-023-06615-2

S. Deacon et al

Diagnostic Histopathology, December 2024 (30), Issue 12, 691 - 698

Respiratory Metagenomics + Pandemic Surveillance

Jonathan Edgeworth (Guy's and St Thomas' NHS Trust)



[Home](#) > [Health and social care](#) > [Public health](#) > [Health protection](#)

Press release

UK to create world-first 'early warning system' for pandemics

The government is set to partner with Oxford Nanopore, which uses technology to rapidly diagnose a range of cancers, along with rare and infectious diseases.

From: [Department of Health and Social Care](#), [Department for Science, Innovation and Technology](#), [UK Health Security Agency](#) and [The Rt Hon Wes Streeting MP](#)

Published 5 November 2024

Routine Metagenomics Service for ICU Patients with Respiratory Infection

Themoula Charalampous¹, Adela Alcolea-Medina^{1 2}, Luke B Snell^{1 3}, Christopher Alder^{1 3}, Mark Tan¹, Tom G S Williams³, Noor Al-Yaakoubi¹, Gul Humayun¹, Christopher I S Meadows^{4 5}, Duncan L A Wyncoll⁵, Richard Paul⁵, Carolyn J Hemsley³, Dakshika Jeyaratnam³, William Newsholme³, Simon Goldenberg³, Amita Patel^{1 3}, Fearghal Tucker², Gaia Nebbia³, Mark Wilks⁶, Meera Chand⁷, Penelope R Cliff², Rahul Batra^{1 3}, Justin O'Grady⁸, Nicholas A Barrett⁵, Jonathan D Edgeworth^{1 3}

Affiliations + expand

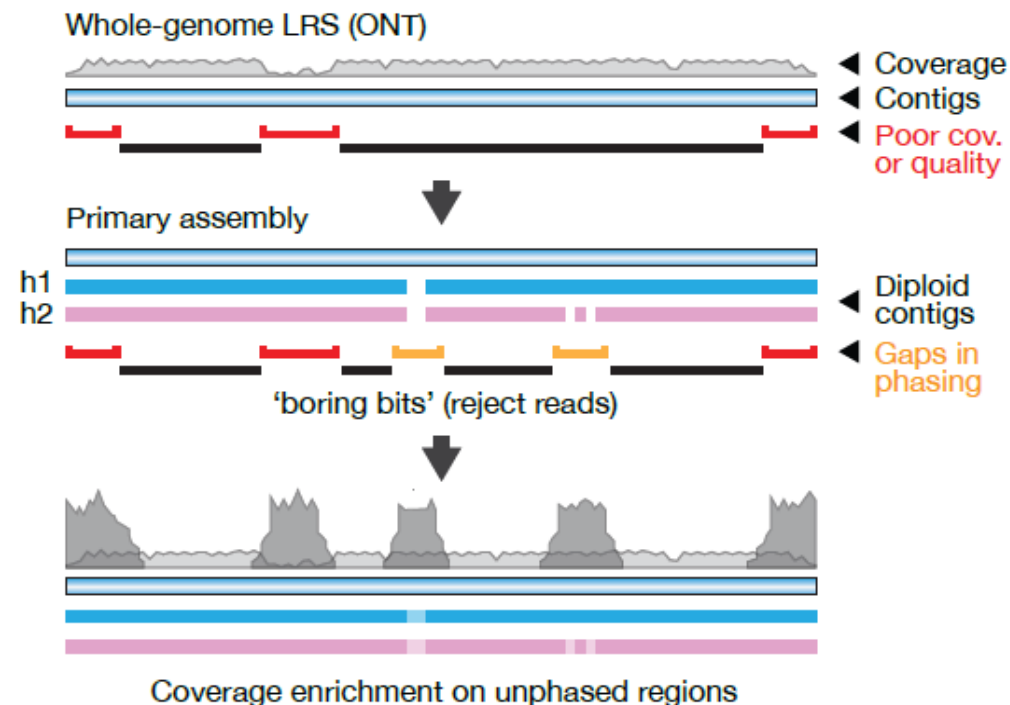
PMID: 37938162 PMCID: PMC10806431 DOI: 10.1164/rccm.202305-0901OC

Abstract

Rationale: Respiratory metagenomics (RMg) needs evaluation in a pilot service setting to determine utility and inform implementation into routine clinical practice. **Objectives:** Feasibility, performance,

Dynamic Adaptive Sampling: Cornetto

Hasindu Gamaarachchi (Garvan Institute)



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New Results

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Adaptively integrated sequencing and assembly of near-complete genomes

 Hasindu Gamaarachchi,  Igor Stevanovski, Jillian M. Hammond, Andre L. M. Reis, Melissa Rapadas, Kavindu Jayasooriya, Tonia Russell, Dennis Yeow, Yvonne Hort, Andrew J. Mallett, Elaine Stackpoole, Lauren Roman, Luke W. Silver,  Carolyn J. Hogg, Lou Streeting,  Ozren Bogdanovic, Renata Coelho Rodrigues Noronha, Luis Adriano Santos do Nascimento, Adauto Lima Cardoso, Arthur Georges, Haoyu Cheng,  Hardip R. Patel,  Kishore Raj Kumar, Amali C. Mallawaarachchi,  Ira W. Deveson

doi: <https://doi.org/10.1101/2025.03.31.646505>



Towards a truly multiomic platform



Chris Seymour ✓

@iiSeymour

...



12:26 am · 27 Nov 2020

Exquisite Sensitivity

Neutron-level resolution

Article | Published: 12 September 2010

A primary hydrogen–deuterium isotope effect observed at the single-molecule level

[Siran Lu](#), [Wen-Wu Li](#), [Dvir Rotem](#), [Ellina Mikhailova](#) & [Hagan Bayley](#) 

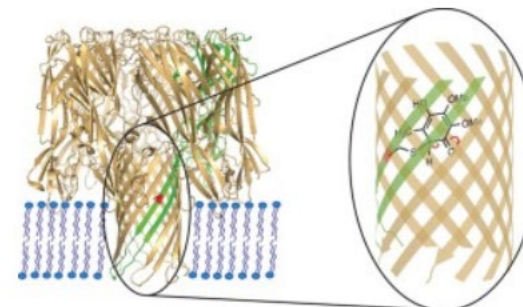
Nature Chemistry **2**, 921–928 (2010) | [Cite this article](#)

1816 Accesses | 1 Altmetric | [Metrics](#)

Abstract

The covalent chemistry of reactants tethered within a single protein pore can be monitored by observing the time-dependence of ionic current flow through the pore, which responds to bond making and breaking in individual reactant molecules. Here we use this ‘nanoreactor’ approach to examine the reaction of a quinone with a thiol to form a substituted hydroquinone by reductive 1,4-Michael addition. Remarkably, a primary hydrogen–deuterium isotope effect is readily detected at the single-molecule level during prototropic rearrangement of an initial adduct. The observation of individual reaction intermediates

Figure 1: A pore-forming protein acts as a reactor that allows measurement of kinetics on a single molecule.



Exquisite Sensitivity

Neutron-level resolution



Computational and Structural Biotechnology
Journal

Volume 23, December 2024, Pages 3584-3594



Research Article

Nanopore sequencing enables novel detection of deuterium incorporation in DNA

Christian Höner zu Siederdisen ^a , Janes Spangenberg ^a, Kevin Bisdorf ^b,
Sebastian Krautwurst ^a, Akash Srivastava ^a, Manja Marz ^{a d e f}, Martin Taubert ^{b c}

Show more

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<https://doi.org/10.1016/j.csbj.2024.09.027>

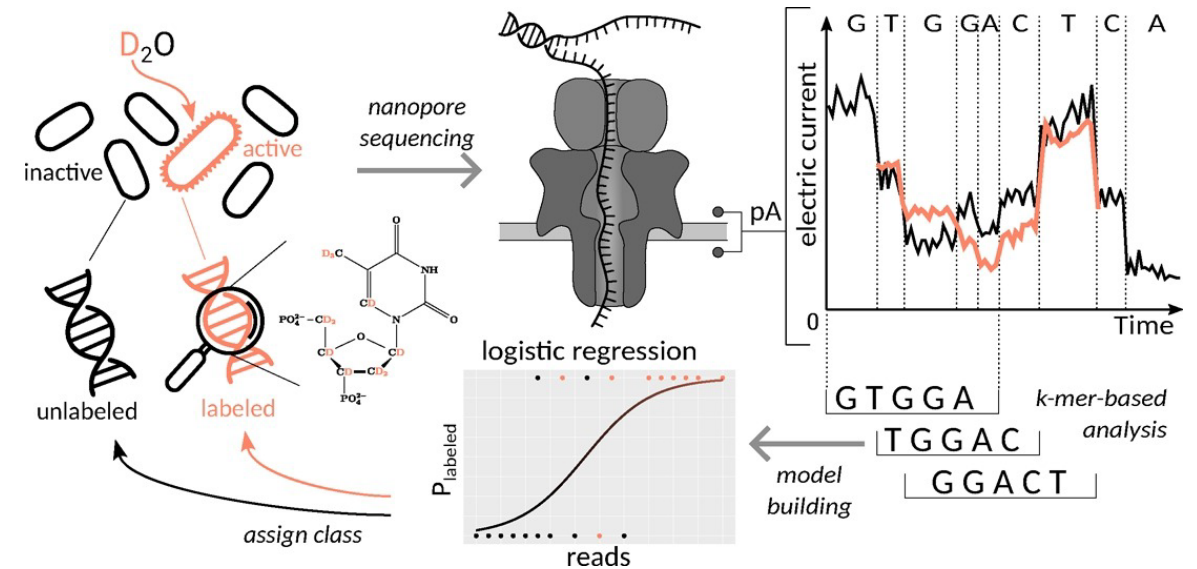
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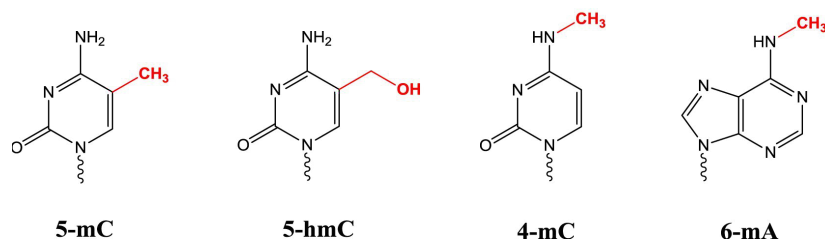
open access

Abstract

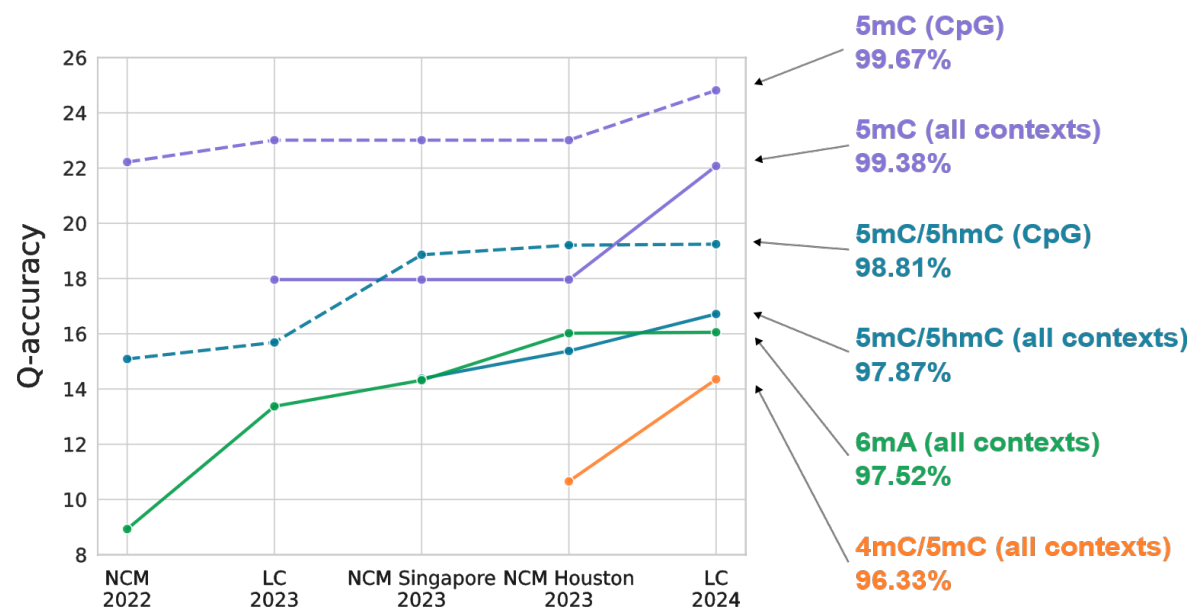
Identifying active microbes is crucial to understand their role in ecosystem functions. Metabolic labeling with heavy, non-radioactive isotopes, i.e., stable isotope probing (SIP),



Current DNA modifications included

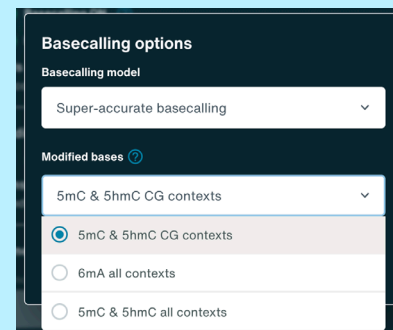


Evolution of Remora model accuracy



Raw read modification accuracy

Fully integrated in **MinKNOW**



Available live or post-run

- 5mC and 5hmC – “CpG”
- 5mC and 5hmC all contexts
- 6mA – all contexts

Offline offering – Dorado v0.7.0

DNA all contexts: 5mC, 5hmC, 6mA and 4mC

```
dorado basecaller hac,5mC_5hmC,6mA > basecalls.bam
```

Enables the use of 6mA footprinting
for chromatin accessibility

Towards detection of all DNA damaged bases

Genomic uracil

- Deoxy-uracil found genomic DNA at low level
- Elevated in some pathologic conditions
- Used by chemotherapy to trigger apoptosis



Next DNA adducts in research

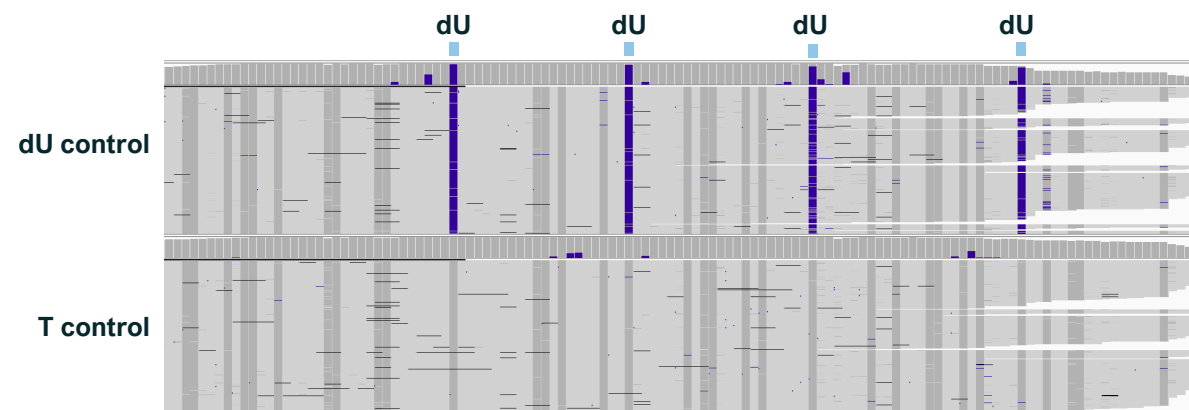
- Oxidation G > 8oxoG
- Ribo-bases incorporation

Highly accurate dU research model

True labels	T	dU
	99.54	0.46
dU	6.85	93.15
Predicted labels		

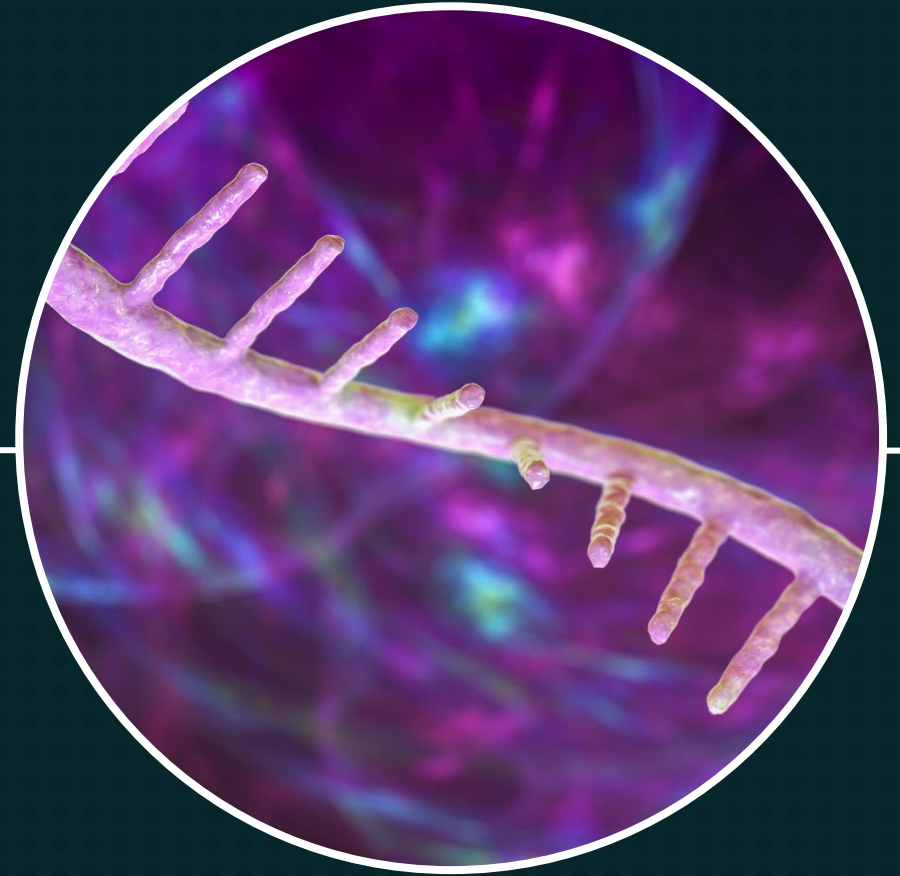
Accuracy: 96.34%

Confusion matrix of pass calls obtained from dU calling on synthetic oligonucleotides with custom dU Dorado v4.2 SUP model



Validation results obtained from dU calling on synthetic oligonucleotides with custom dU Dorado v4.2 SUP model. Positions of modifications on the top track, followed by the modification calls from Dorado for dU. Top read track: dU modified control, bottom read track: unmodified control. In total we used 5 test oligos with 31 dU sites.

RNA



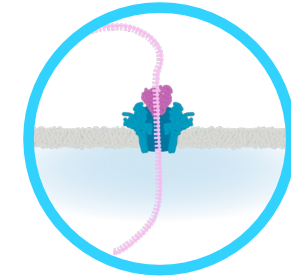
Direct RNA sequencing

Available now

Significant changes released in 2023 with new RNA004 chemistry



- Native RNA with modified bases sequenced
- RNA specific pore and motor protein (3' down)
- Poly(A) tail length estimation
- Quick library preparations (30–120 mins)
- 98.8% median accuracy per read



RNA004

RNA001

RNA002

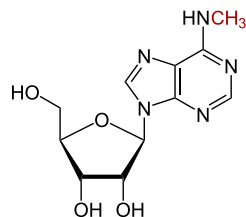
2013 2014 2015 2016 **2017** 2018 2019 2020 2021 2022 2023 2024



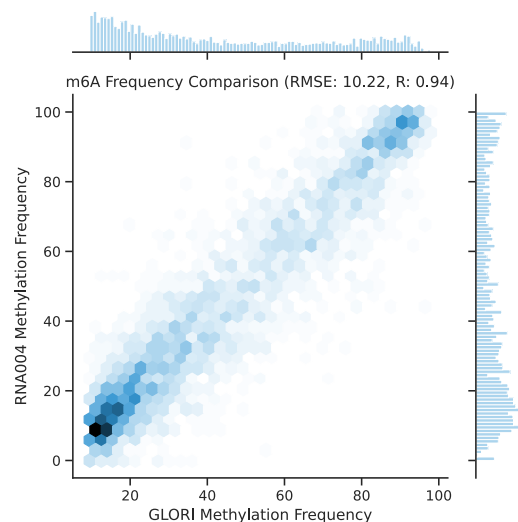
The only method for directly sequencing native RNA released

Direct RNA modified bases - SOTA

m6A



m6A detection correlation with GLORI

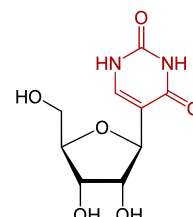


True labels	Predicted labels	
	A	m6A
A	97.64	2.36
m6A	3.40	96.60

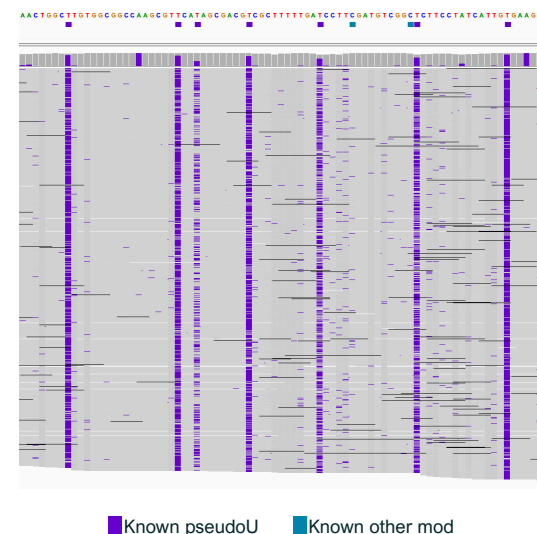
Single base comparison of the level of m6A methylation in DRACH context of WT HEK-293 cells between RNA004 data and a matched GLORI sequencing dataset obtained from Liu C et al. Nat Biotechnol. 2023;41(3):355-366.

Accuracy 97.12%
(DRACH only : 99.17%)

PseudoU



PseudoU detection at known rRNA sites



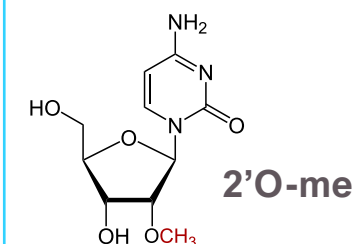
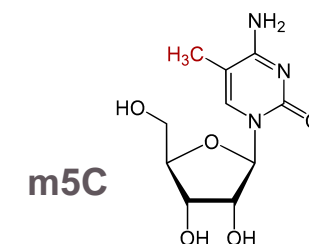
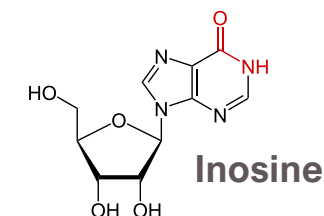
True labels	Predicted labels	
	U	pseU
U	99.15	0.85
pseU	3.90	96.10

IGV screenshot of rRNA 28S (4,369-4,445) from UHRR control RNA004 datasets produced with rRNA specific adapters. PseudoU calling performed with Dorado RNA004 v5 + PseU v1 models. Data courtesy of M. Jain. Genome Technology Laboratory, Northeastern University and M. Akeson. Deamer Laboratory, UCSC

Accuracy 97.62%

Next...

More mods in Research



Now in Dorado v0.7.0

Transfer RNA – ‘Epitransferomics’

Also with modifications

► bioRxiv [Preprint]. 2024 Jul 24:2024.07.23.604651. [Version 1] doi: [10.1101/2024.07.23.604651](https://doi.org/10.1101/2024.07.23.604651)

Comparative analysis of 43 distinct RNA modifications by nanopore tRNA sequencing

Laura K White¹, Kezia Dobson¹, Samantha del Pozo¹, Jill M Bilodeaux¹, Shelby E Andersen², Amber Baldwin¹, Chloe Barrington¹, Nadine Körtel¹, Federico Martinez-Seidel¹, Saylor M Strugar¹, Kristin EN Watt³, Neelanjan Mukherjee¹, Jay R Hesselberth^{1,+}

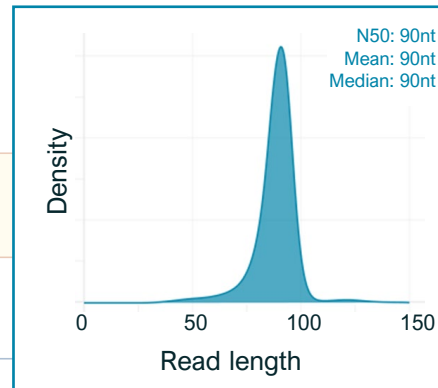
► Author information ► Copyright and License information

PMCID: PMC11291079 PMID: [39091754](https://pubmed.ncbi.nlm.nih.gov/39091754/)

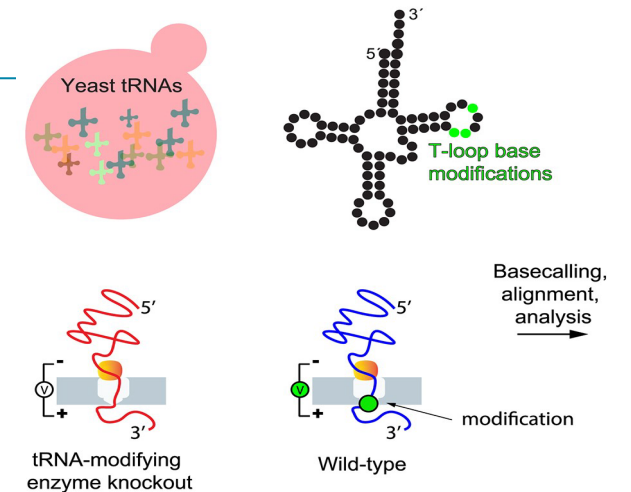
The complete version history of this preprint is available at [bioRxiv](https://www.biorxiv.org/content/10.1101/2024.07.23.604651v1).

Abstract

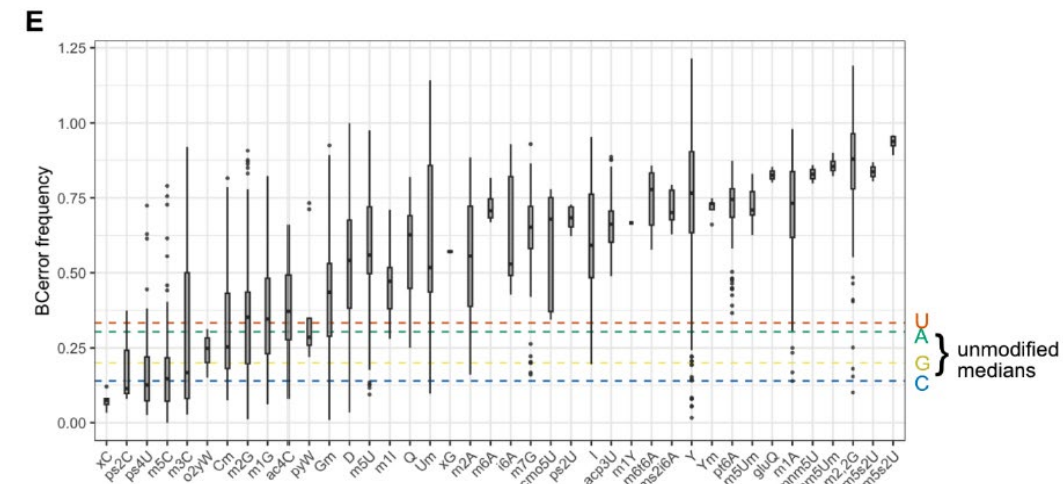
Transfer RNAs are the fundamental adapter molecules of protein synthesis and the most abundant and heterogeneous class of noncoding RNA molecules in cells. The study of tRNA repertoires remains challenging, complicated by the presence of dozens of post transcriptional modifications. Nanopore sequencing is an emerging technology with promise



Nanopore direct RNA sequencing



Nucleic Acids Res, Volume 52, Issue 19, 28 October 2024, Pages 12074–12092, <https://doi.org/10.1093/nar/gkae796>

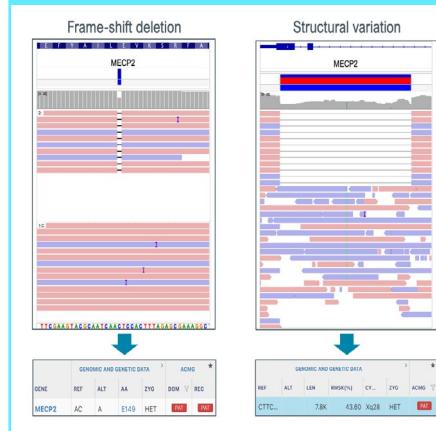


Enabling 'multiomics' for human biology

On a single platform

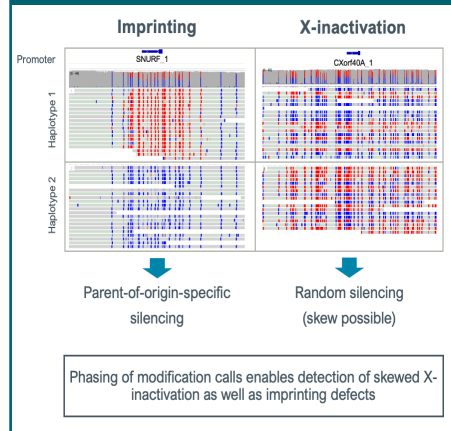
Phasing

Canonical bases



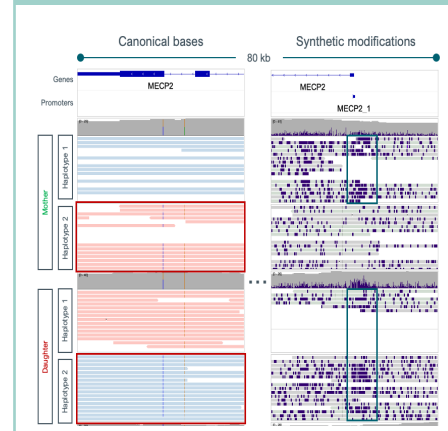
- > Full spectrum of variants (SNVs, SVs, CNVs)
- > Haplotype resolved
- > Including hard to analyze regions

Native modifications



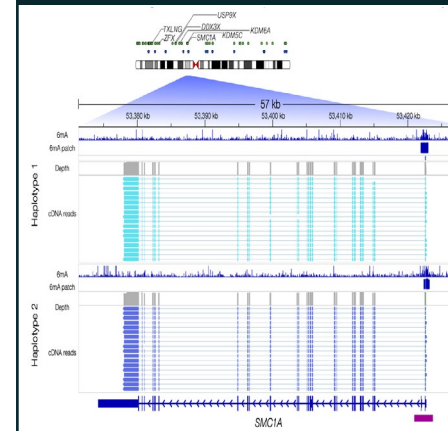
- > 5mC and 5hmC included
- > Imprinting disorders
- > Methylation signatures

Synthetic modifications



- > Mark region of interest with 6mA
- > Nucleosome spacing
- > DiMeLo-seq

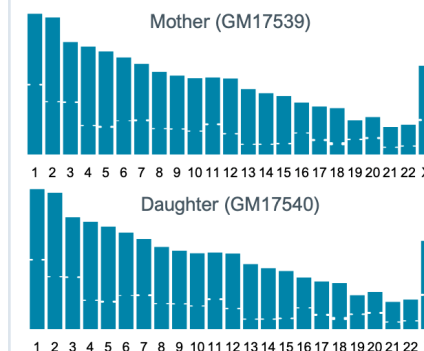
RNA



- > Expression and splicing
- > Transposable elements

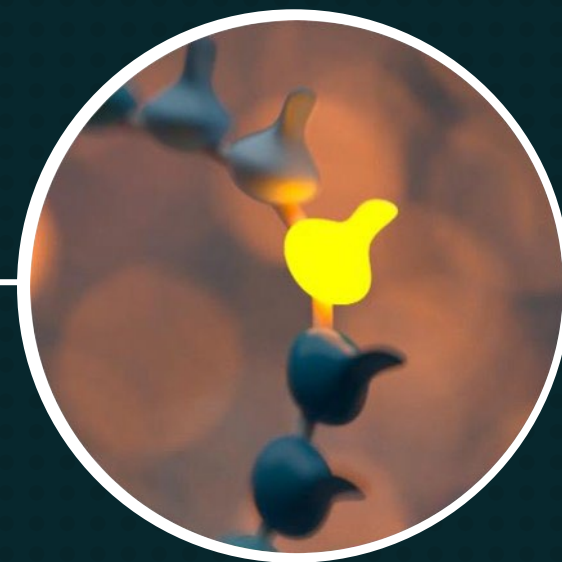
Chromatin confirmation

Pore-C phasing



- > Enhancers interactions
- > X chromatin structure

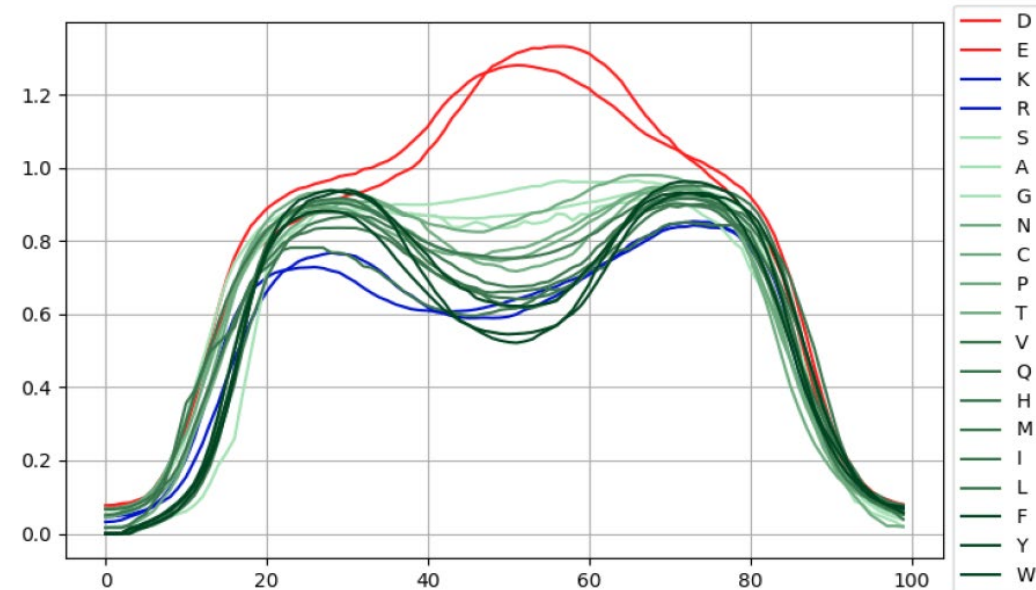
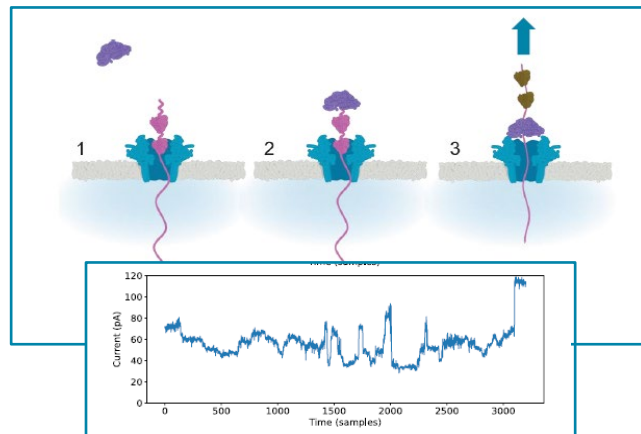
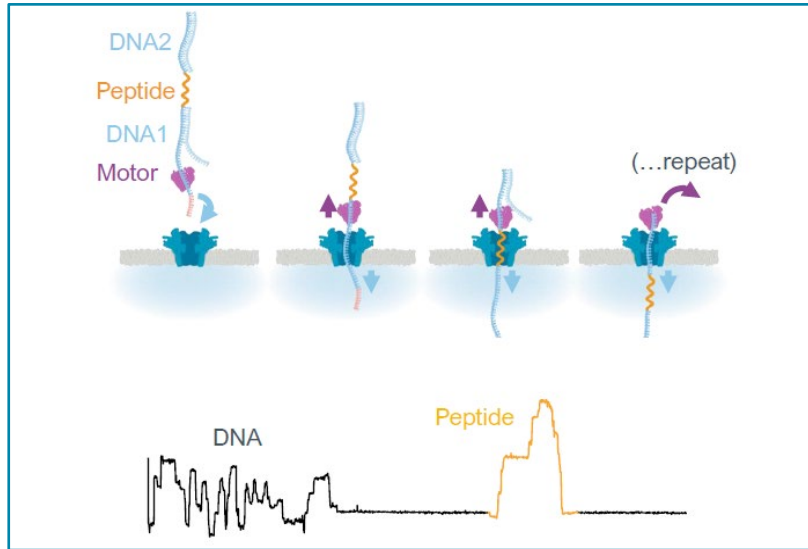
AA



Peptides and Proteins



More signal complexity



Article

Multi-pass, single-molecule nanopore reading of long protein strands

<https://doi.org/10.1038/s41586-024-07935-7>

Received: 28 June 2023

Accepted: 9 August 2024

Keisuke Motone^{1,4,5}, Daphne Kontogiorgos-Heintz^{1,5}, Jasmine Wee¹, Kyoko Kurihara¹, Sangbeom Yang¹, Gwendolin Roote¹, Oren E. Fox¹, Yishu Fang¹, Melissa Queen¹, Mattias Tolhurst², Nicolas Cardozo², Miten Jain³ & Jeff Nivala^{1,2,✉}

Scaling



Pushing scale for human genomics

The 1000 Genomes Project

3,000 sample program

407 SVs in coding regions
unreliable for short-read
sequencing

1KGPONT
Sequencing
Consortium

PRECISE Singapore

10,000 genomes
representing Singapore's
diverse population



Genomics England & UKBB

Cancer 2.0, rare disease
on over 7,500 samples



NIH CARD

4,000 brain samples
uncovering novel insights
into Parkinson's,
Alzheimer's and more



NIHR Bioresource

22,000 genomes
representing rare disease
patients



Pan Arab Genome

1M base pairs of novel
euchromatic sequences



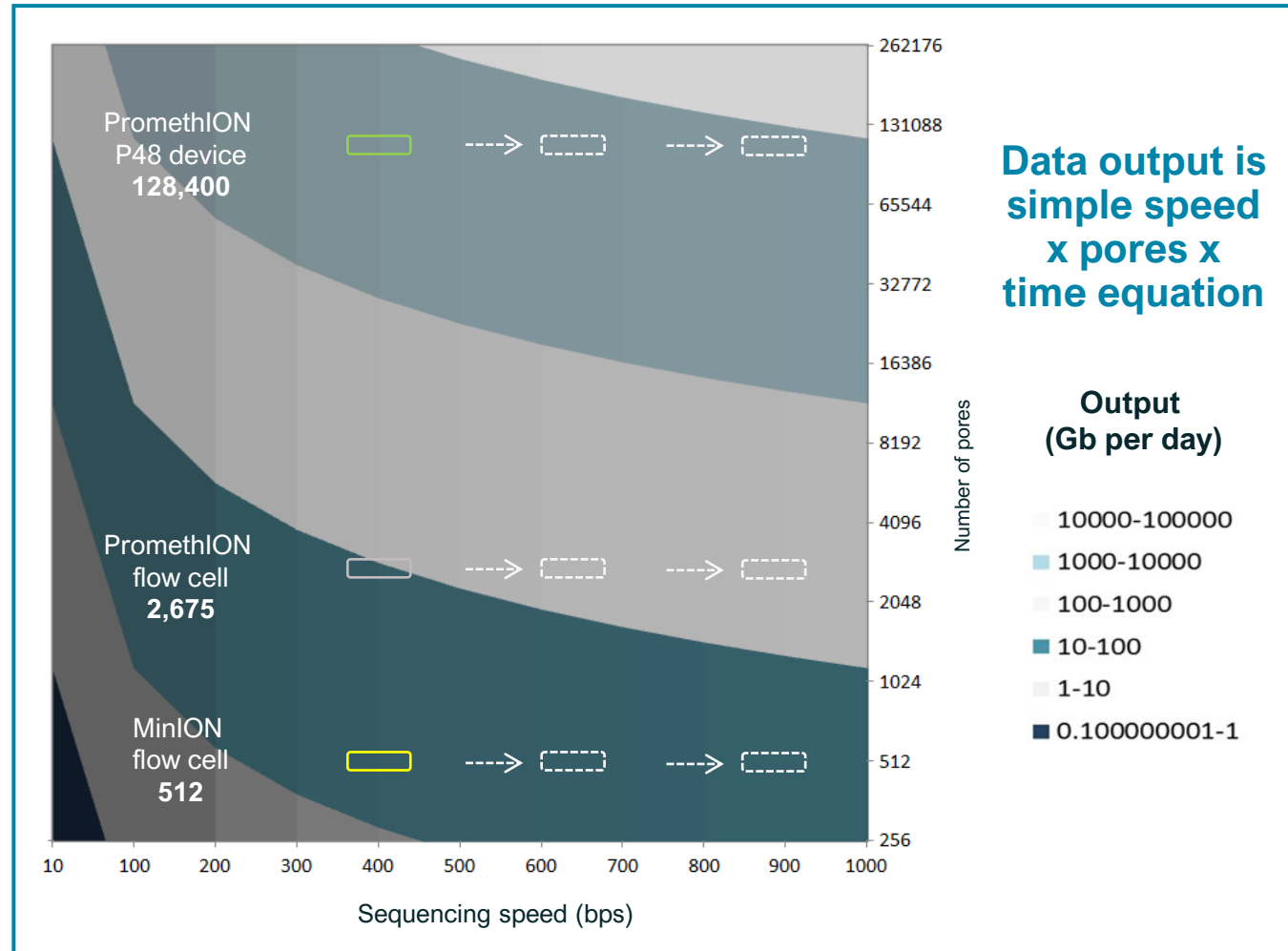
Emirati Genome Project

Over 100,000
genomes sequenced



Scaling nanopore data output

Towards **15-minute** genomes



AGBT 2012: Full circle

High end example: 20-node installation to deliver human genome in 15 minutes if required

- GridION 8k cartridge (available 2013)
- Nanopore speed: ~300b/s
- 20 node installation
- Total system: 3Tb per day
- One genome in 15 minutes with no requirement to pool sample
- ~50 fold coverage
- <\$10 per Gb (all inclusive)

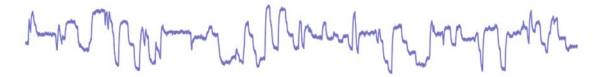
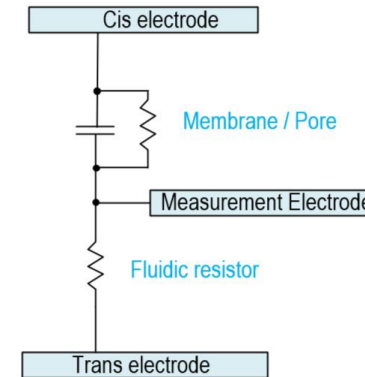


Voltage sensor

Route to future 100k channel sequencing chip

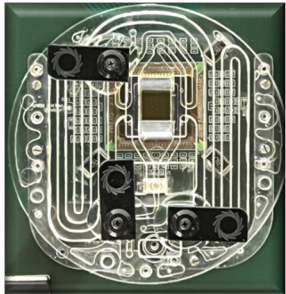
Same chemistry radically different chip design

- Nanopore density on current flow cell is limited by the fluidic well
- Switching to voltage sensor allows us to increase the well density by $\sim \times 10$.
- Target = 100k channel flow cell

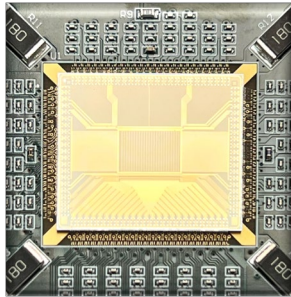


Output signal (Voltage)

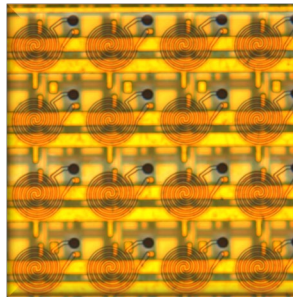
Progress so far



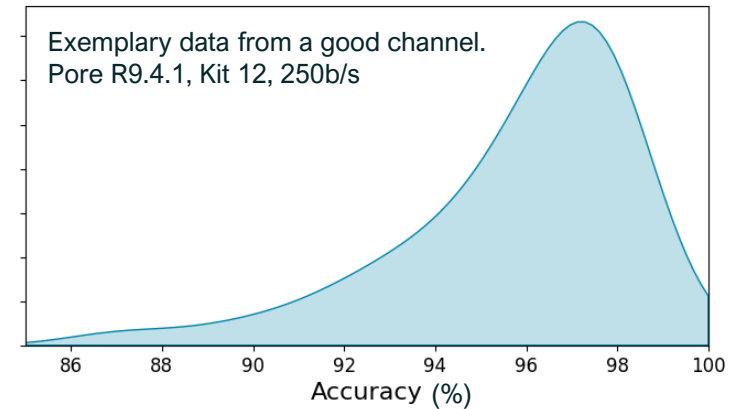
Flowcell



ASIC



High density array with MEMS fluidic resistor



- Specialised voltage sensor ASIC designed and fabricated
- Test chips and flow cells designed and built

- 100k reads obtained and basecaller trained
- Achieving median 97% raw read accuracy

What and Where next?

In the clinic, in the field, at home...

Healthcare



Viral vs Bacteria test by G.P or at home

Cancer surgery resection with real time feedback during operation

Dentists using devices for rapid tooth infection analysis prior to treatment

Food and Security



Farmers determining diseased crops or root stock before planting

‘DNA Certified’ movement to ensure quality or authenticity, akin to the Organic Food movement

Remote samplers in waterways/taps, sniffers at airports/underground trains continually testing for pathogens

Remote soil based sensors to continually analyse microbiome of fields/crops

Consumer



Telomere testing at home – age/health/longevity tracking (plus methylation)

Daily gene expression tracking and trending, in combination with exercise machines/bikes

Smart toilets or toothbrushes for early cancer detection

Fridge sensors for food spoilage

Thank You

Acknowledgements:

Colleagues, Collaborators, Customers...

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